Coding vs non-coding signals in genomic language model embeddings

Methods and results · 15 October 2026

Scope. This report describes four experiments, and a training-set size robustness analysis, that probe what frozen embeddings of genomic language models (gLMs) encode about protein-coding versus non-coding DNA, and lists their results. Each experiment was run on two kinds of data: the 200 bp Genomic Benchmarks coding vs intergenomic windows, and full-length human sequences (complete coding sequences against intergenic, lncRNA and intron sequences). Twenty-one models from six families were evaluated. The report gives methods, rationale and results only. Interpretation and discussion are left out.

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1 Models

All models are used frozen, with no fine-tuning. Table 1 lists them. The *window* column is the longest stretch of DNA embedded in one forward pass; longer sequences are split into chunks (Section 2.2).

Table 1: Models evaluated. Embedding width is the size of the last hidden layer.

Model	Hugging Face ID	Architecture / tokens	Training genomes	Width	Window (bp)
GROVER	PoetschLab/GROVER	BERT, BPE	human	768	1,500
DNABERT-2	zhihan1996/ DNABERT-2-117M	BERT + ALiBi, BPE	multi-species	768	1,500
NT-v2 50M	InstaDeepAI/nucleotide-transformer-v2-50m-multi-species	ESM-style, 6-mer	multi-species	512	10,000
NT-v2 100M	…-v2-100m-multi-species	ESM-style, 6-mer	multi-species	512	10,000
NT-v2 250M	…-v2-250m-multi-species	ESM-style, 6-mer	multi-species	768	10,000
NT-v2 500M	…-v2-500m-multi-species	ESM-style, 6-mer	multi-species	1024	10,000
Hyena tiny-1k d128	LongSafari/hyenadna-tiny-1k-seqlen-hf	Hyena, single nucleotide	human	128	1,000
Hyena tiny-1k d256	…/hyenadna-tiny-1k-seqlen-d256-hf	Hyena, single nucleotide	human	256	1,000
Hyena tiny-16k	…/hyenadna-tiny-16k-seqlen-d128-hf	Hyena, single nucleotide	human	128	10,000
Hyena small-32k	…/hyenadna-small-32k-seqlen-hf	Hyena, single nucleotide	human	256	10,000
Hyena medium-160k	…/hyenadna-medium-160k-seqlen-hf	Hyena, single nucleotide	human	256	10,000
Hyena medium-450k	…/hyenadna-medium-450k-seqlen-hf	Hyena, single nucleotide	human	256	10,000
Hyena large-1m	…/hyenadna-large-1m-seqlen-hf	Hyena, single nucleotide	human	256	10,000
Caduceus-Ph 1k d118	kuleshov-group/caduceus-ph_seqlen-1k_d_model-118_n_layer-4_lr-8e-3	BiMamba, single nucleotide	human	118	1,000
Caduceus-Ps 1k d118	…/caduceus-ps_seqlen-1k_d_model-118_n_layer-4_lr-8e-3	BiMamba RC-equivariant, single nucleotide	human	236	1,000
Caduceus-Ph 1k d256	…/caduceus-ph_seqlen-1k_d_model-256_n_layer-4_lr-8e-3	BiMamba, single nucleotide	human	256	1,000
Caduceus-Ps 1k d256	…/caduceus-ps_seqlen-1k_d_model-256_n_layer-4_lr-8e-3	BiMamba RC-equivariant, single nucleotide	human	512	1,000
Caduceus-Ph 131k	…/caduceus-ph_seqlen-131k_d_model-256_n_layer-16	BiMamba, single nucleotide	human	256	10,000
Caduceus-Ps 131k	…/caduceus-ps_seqlen-131k_d_model-256_n_layer-16	BiMamba RC-equivariant, single nucleotide	human	512	10,000
Evo 1	togethercomputer/evo-1-8k-base (`evo-1-8k-base`)	StripedHyena 7B, single nucleotide	prokaryotes and phages	4096	4,000
Evo 2	arcinstitute/evo2_7b (`evo2_7b`)	StripedHyena 2 7B, single nucleotide	all domains of life	4096	2,000

Caduceus-Ph and -Ps are the two Caduceus variants: Ph is trained with reverse-complement data augmentation, while Ps is reverse-complement equivariant by construction. Its hidden state holds a forward and a reverse-complement half, so its embedding width is twice the model dimension. The two halves are used as they are, without averaging.

2 Common methodology

2.1 Sequence embeddings

For each sequence the last hidden layer of the model is taken and mean-pooled over tokens, using the attention mask so that padding tokens are excluded. This is the same pooling as in the repository's attribution notebook. The result is one fixed-length vector per sequence, of the width shown in Table 1.

Model-specific details, needed for reproduction:

- **GROVER** and the **HyenaDNA** models load with `AutoModel` (transformers ≥ 5.18).
- **DNABERT-2** needs transformers 4.44–4.46. Its config is passed explicitly as a `BertConfig`, and its Triton flash-attention kernel is disabled so the PyTorch attention path is used.
- **NT-v2** needs the same older transformers. It loads with `AutoModelForMaskedLM` with `output_hidden_states=True`, and the last hidden state is taken.
- **HyenaDNA** and **Caduceus** take no attention mask. Chunks of different lengths are therefore never batched together, so no padding enters their hidden states (causal for HyenaDNA, bidirectional for Caduceus). Caduceus needs transformers 4.46 with `mamba-ssm` 2.2.5 and `causal-conv1d` 1.5.2 and loads with `AutoModel`.
- **Evo 1** and **Evo 2** load through their own packages (`evo-model`, `evo2`). Embeddings are the output of the final normalisation layer (the last hidden layer), captured with a forward hook and mean-pooled. Tokens are single characters with no special tokens, and only equal-length chunks are batched, so there is no padding.

2.2 Long sequences: chunk and average

Full-length sequences (300 bp to 10 kb) are often longer than a model's context. A sequence of length L is split into $\lceil L/W \rceil$ contiguous chunks of near-equal length, where W is the model's window (Table 1). Each chunk is embedded, and the sequence embedding is the mean of the chunk embeddings, weighted by chunk length.

The windows were chosen as follows:

- GROVER and DNABERT-2 are limited to 512 tokens. 1.5 kb stays below that even for poorly compressible sequence; 2 kb reached about 550 tokens.
- The HyenaDNA and Caduceus 1k models use their 1 kb training length.
- Evo 1 (4 kb) and Evo 2 (2 kb) are limited by GPU memory on a shared 24 GB A10, not by context length. Evo 1 at 8 kb ran out of memory.
- The other models use 10 kb, so a sequence is never split for them.

Every 200 bp window is a single chunk, so in that case the embedding is the plain per-sequence embedding.

2.3 Linear probe and coding score

Every experiment uses the same probe:

- Standardisation followed by L2-regularised logistic regression (scikit-learn `LogisticRegression`, `C = 1`, `max_iter = 2000`).
- Trained on the training-set embeddings with labels 0 = coding and 1 = non-coding.
- The *coding score* of a sequence is the negative of the probe's decision function, so higher means more coding-like. A sequence is *called coding* when its coding score is above 0.

Experiments B–D keep the probe fixed and only change the input sequences, so any change in the coding score comes from the model's response to the edit.

2.4 Metrics

- 200 bp data: accuracy, because the test set is balanced.
- Full-length data: balanced accuracy (the mean of per-class recall). The chromosome-held-out test sets are unbalanced (Table 2), and with plain accuracy chance level would differ between datasets (up to 0.585).

2.5 Software

- Analyses are in `Model Evaluation/Coding-Noncoding/Embedding-Patterns/`.
- They run under pixi with five environments: `default` (transformers 5; GROVER, HyenaDNA), `legacy` (transformers 4.44–4.46; DNABERT-2, NT-v2), `caduceus`, `evo` and `evo2` (PyPI torch 2.7.1 with prebuilt CUDA kernels).

- Hardware: NVIDIA A10 GPUs (24 GB).

3 Datasets

3.1 200 bp: Genomic Benchmarks

- Data set: `katarinagresova/Genomic_Benchmarks_demo_coding_vs_intergenomic_seqs`, 200 bp human windows labelled coding or intergenomic.
- Train: 75,000 windows (37,500 per class). Test: 25,000 windows (12,500 per class).
- The original split is used.

The reading frame of the coding windows is not given. Experiments B and C need it, so it is inferred: a window is assigned the frame if exactly one of its three forward frames has no stop codon. This holds for 8,824 of the 12,500 coding test windows (frames 0/1/2: 2,942/2,936/2,946). The other 3,676 windows have no reading frame assigned and are excluded wherever a frame is needed.

3.2 Full-length human sequences

The 200 bp windows are short (about 66 codons), cut out of genes, and not aligned to the reading frame. Complete coding sequences remove these limits, and having several kinds of non-coding DNA separates "coding vs non-coding" from "coding vs one particular background".

All sets come from Ensembl GRCh38 release 116 (`prepare_full_length.py`). Rules common to every set:

- Sequences are 300 bp to 10 kb long and contain only A/C/G/T.
- One sequence per gene.
- Chromosomes 1–22, X and Y only.

Set	Construction
coding	CDS of the `Ensembl_canonical` transcript of each protein-coding gene. It must start with ATG, end with a stop codon, have a length divisible by 3 and no internal in-frame stop codon. Reading frame 0 is known for every sequence.
lncrna	Canonical transcript of each lncRNA gene (spliced, from the Ensembl ncRNA FASTA).
intron	One randomly chosen intron per canonical protein-coding transcript, taken from the gaps between GTF exons. It is reverse-complemented on minus-strand genes, so it reads in the sense direction.
intergenic	Random genome segments at least 10 kb from any annotated gene of any biotype. They are sampled with exactly the lengths drawn from the coding length distribution, and must contain no N.

Length matching.

- Sequence length strongly affects composition statistics and chunking, so each non-coding set is sampled to follow the coding length distribution.
- lncRNA and intron sequences are matched in 20 log-spaced length bins. Intergenic segments copy coding lengths directly.
- Table 2 gives the Kolmogorov–Smirnov statistic against the coding lengths, and Figure 1 shows the distributions.

Train/test split by chromosome. Sequences on chromosomes 2, 4, 6 and 8 form the test set; all others form the training set. A random split would let paralogous genes, or a gene's CDS and its own intron or neighbouring lncRNA, appear in both training and test sets.

Datasets. The three binary datasets `cds_vs_intergenic`, `cds_vs_lncrna` and `cds_vs_intron` share the same coding set. In each, the probe is trained on the coding training sequences against that non-coding set's training sequences.

Table 2: Full-length sequence sets. GC is the mean per-sequence GC fraction.

set	n	train	test	median length	length IQR	mean GC	KS D vs coding
coding	10000	8165	1835	1323	861-2085	0.530	0.000
intergenic	10000	7410	2590	1311	843-2052	0.385	0.011
lncrna	9842	7533	2309	1292	854-2034	0.471	0.020
intron	9862	8054	1808	1333	858-2098	0.467	0.012

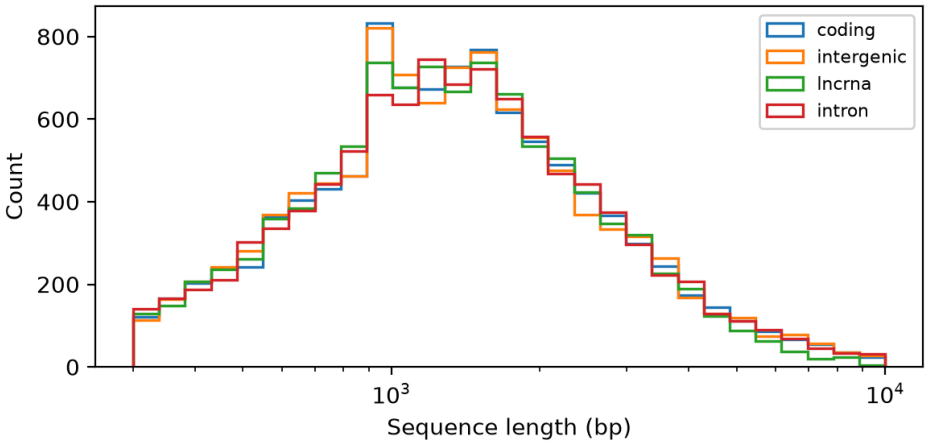


Figure 1: Length distributions of the full-length sets.

4 Experiment A: k-mer baseline and residual probe

4.1 Rationale

Coding and non-coding DNA differ strongly in base composition (GC content, codon-driven k-mer frequencies). A probe on gLM embeddings may separate the classes only because the embeddings encode composition. Experiment A measures how far composition alone gets, and whether anything useful remains in the embeddings once composition is removed.

4.2 Method

1. **k-mer probes.**
   - For each $k = 1, \ldots, 6$, every sequence is represented by the frequencies of its overlapping k-mers (counts divided by the total, so sequences of different lengths are comparable).
   - A logistic regression (scaled, `max_iter = 2000`) is trained on these frequencies.
   - k-mer probes do not involve a gLM, so they are the same for every model.
2. **Embedding probe.** The standard probe on the gLM embeddings.
3. **Residual probe.**
   - The 1- to 6-mer frequencies are concatenated (5,460 features) and standardised.
   - A multi-output ridge regression ($\alpha = 10$) is fitted on the training set to predict the embedding from them.
   - The prediction is subtracted from the embedding for both training and test sequences, and the standard probe is trained and tested on these residuals.
   - The residual is the part of the embedding that k-mer frequencies cannot linearly explain. Residual accuracy above chance (0.5) means that part still carries class information.

4.3 Results

Table 3: k-mer frequency probes, 200 bp Genomic Benchmarks (accuracy).

	1-mer	2-mer	3-mer	4-mer	5-mer	6-mer
k-mer probe	0.758	0.815	0.870	0.881	0.895	0.895

Table 4: Embedding and residual probes, 200 bp Genomic Benchmarks. *dims* = embedding width.

Model	Embedding	Residual	dims
GROVER	0.906	0.592	768
DNABERT-2	0.912	0.574	768
NT-v2 50M	0.893	0.567	512
NT-v2 100M	0.886	0.556	512
NT-v2 250M	0.907	0.577	768
NT-v2 500M	0.892	0.557	1024
Hyena tiny-1k d128	0.866	0.539	128
Hyena tiny-1k d256	0.879	0.547	256
Hyena tiny-16k	0.864	0.546	128
Hyena small-32k	0.879	0.561	256
Hyena medium-160k	0.881	0.564	256
Hyena medium-450k	0.868	0.570	256
Hyena large-1m	0.873	0.558	256
Caduceus-Ph 1k d118	0.874	0.555	118
Caduceus-Ps 1k d118	0.887	0.563	236
Caduceus-Ph 1k d256	0.900	0.579	256
Caduceus-Ps 1k d256	0.911	0.581	512
Caduceus-Ph 131k	0.922	0.614	256
Caduceus-Ps 131k	0.936	0.631	512
Evo 1	0.894	0.611	4096
Evo 2	0.931	0.627	4096

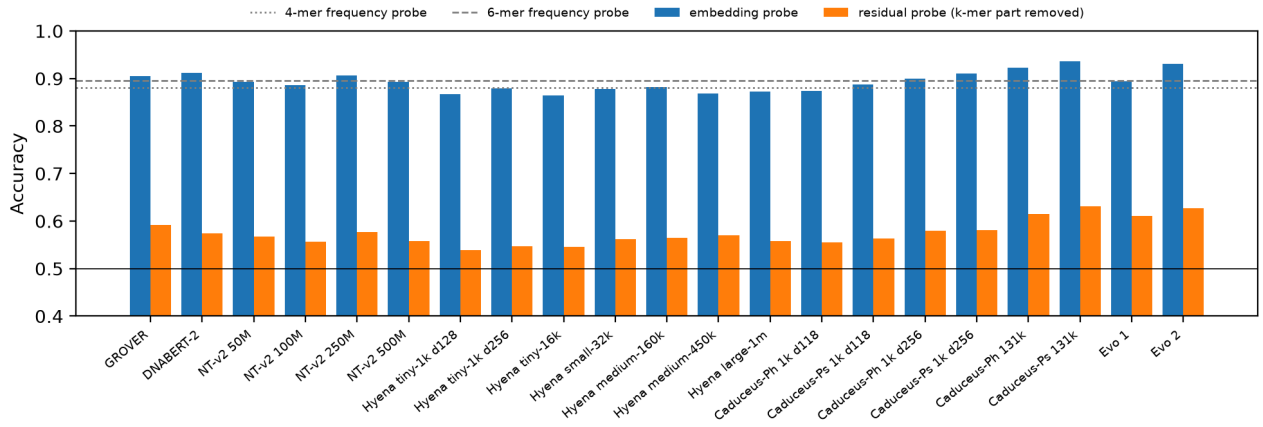


Figure 2: Embedding and residual probe accuracy per model, 200 bp Genomic Benchmarks. Dotted and dashed lines: 4-mer and 6-mer probes.

Table 5: k-mer frequency probes, full length, CDS vs intergenic (balanced accuracy).

	1-mer	2-mer	3-mer	4-mer	5-mer	6-mer
k-mer probe	0.818	0.911	0.972	0.983	0.978	0.987

Table 6: Embedding and residual probes, full length, CDS vs intergenic. *dims* = embedding width.

Model	Embedding	Residual	dims
GROVER	0.988	0.606	768
DNABERT-2	0.994	0.623	768
NT-v2 50M	0.996	0.618	512
NT-v2 100M	0.996	0.609	512
NT-v2 250M	0.998	0.618	768
NT-v2 500M	0.996	0.599	1024
Hyena tiny-1k d128	0.980	0.554	128
Hyena tiny-1k d256	0.979	0.556	256
Hyena tiny-16k	0.977	0.574	128
Hyena small-32k	0.986	0.595	256
Hyena medium-160k	0.979	0.589	256
Hyena medium-450k	0.987	0.582	256
Hyena large-1m	0.982	0.597	256
Caduceus-Ph 1k d118	0.973	0.574	118
Caduceus-Ps 1k d118	0.981	0.572	236
Caduceus-Ph 1k d256	0.989	0.582	256
Caduceus-Ps 1k d256	0.993	0.573	512
Caduceus-Ph 131k	0.993	0.616	256
Caduceus-Ps 131k	0.996	0.628	512
Evo 1	0.991	0.628	4096
Evo 2	0.991	0.613	4096

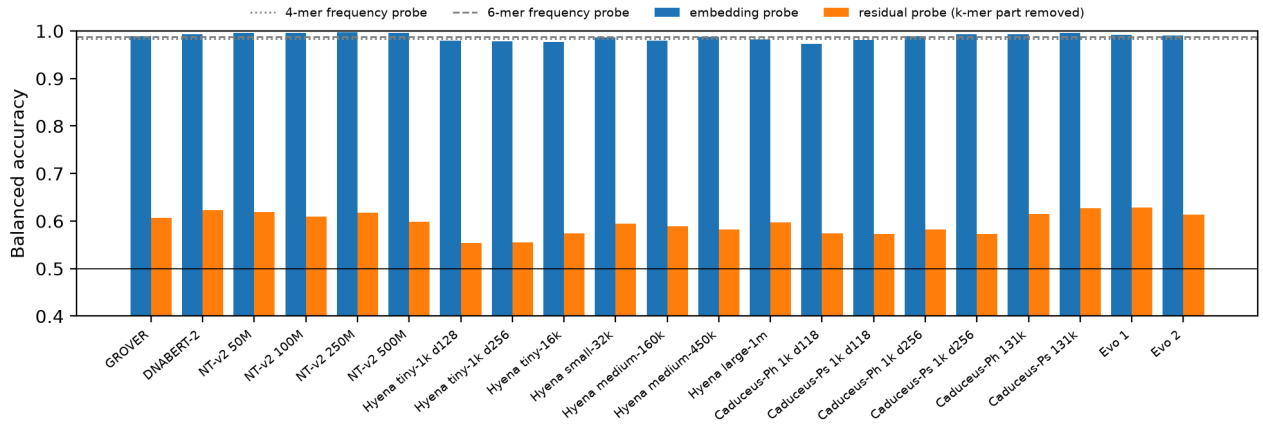


Figure 3: Embedding and residual probe accuracy per model, full length, CDS vs intergenic. Dotted and dashed lines: 4-mer and 6-mer probes.

Table 7: k-mer frequency probes, full length, CDS vs lncRNA (balanced accuracy).

	1-mer	2-mer	3-mer	4-mer	5-mer	6-mer
k-mer probe	0.648	0.814	0.946	0.976	0.968	0.932

Table 8: Embedding and residual probes, full length, CDS vs lncRNA. *dims* = embedding width.

Model	Embedding	Residual	dims
GROVER	0.953	0.605	768
DNABERT-2	0.968	0.656	768
NT-v2 50M	0.993	0.651	512
NT-v2 100M	0.991	0.662	512
NT-v2 250M	0.993	0.657	768
NT-v2 500M	0.991	0.644	1024
Hyena tiny-1k d128	0.927	0.560	128
Hyena tiny-1k d256	0.937	0.558	256
Hyena tiny-16k	0.922	0.572	128
Hyena small-32k	0.956	0.593	256
Hyena medium-160k	0.949	0.618	256
Hyena medium-450k	0.942	0.621	256
Hyena large-1m	0.946	0.620	256
Caduceus-Ph 1k d118	0.936	0.574	118
Caduceus-Ps 1k d118	0.948	0.579	236
Caduceus-Ph 1k d256	0.958	0.589	256
Caduceus-Ps 1k d256	0.969	0.599	512
Caduceus-Ph 131k	0.971	0.654	256
Caduceus-Ps 131k	0.983	0.655	512
Evo 1	0.987	0.673	4096
Evo 2	0.983	0.664	4096

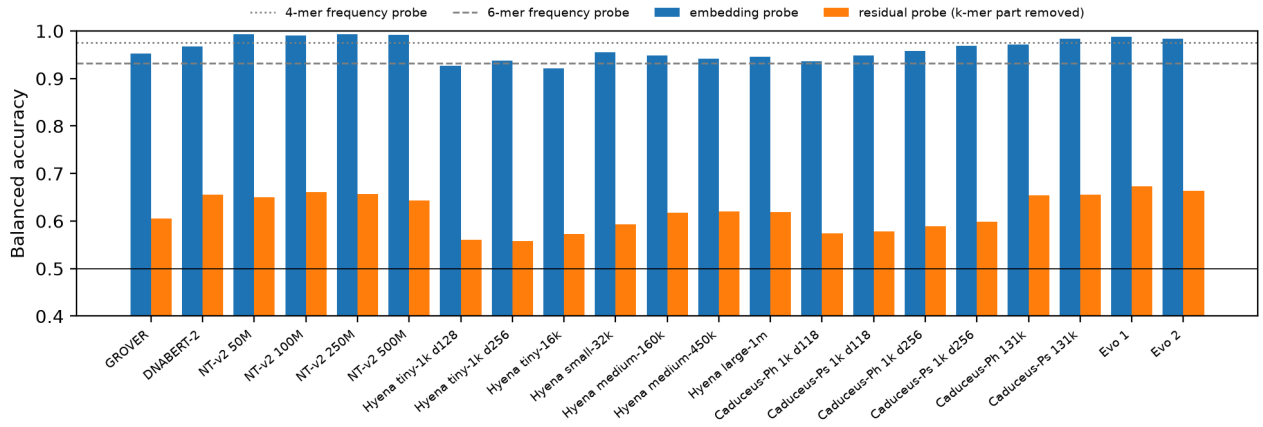


Figure 4: Embedding and residual probe accuracy per model, full length, CDS vs lncRNA. Dotted and dashed lines: 4-mer and 6-mer probes.

Table 9: k-mer frequency probes, full length, CDS vs intron (balanced accuracy).

	1-mer	2-mer	3-mer	4-mer	5-mer	6-mer
k-mer probe	0.806	0.949	0.993	0.994	0.991	0.995

Table 10: Embedding and residual probes, full length, CDS vs intron. *dims* = embedding width.

Model	Embedding	Residual	dims
GROVER	0.990	0.596	768
DNABERT-2	0.996	0.648	768
NT-v2 50M	0.998	0.629	512
NT-v2 100M	0.997	0.622	512
NT-v2 250M	0.999	0.630	768
NT-v2 500M	0.997	0.625	1024
Hyena tiny-1k d128	0.987	0.549	128
Hyena tiny-1k d256	0.988	0.544	256
Hyena tiny-16k	0.979	0.560	128
Hyena small-32k	0.989	0.569	256
Hyena medium-160k	0.987	0.581	256
Hyena medium-450k	0.992	0.595	256
Hyena large-1m	0.989	0.603	256
Caduceus-Ph 1k d118	0.986	0.543	118
Caduceus-Ps 1k d118	0.990	0.556	236
Caduceus-Ph 1k d256	0.992	0.571	256
Caduceus-Ps 1k d256	0.994	0.585	512
Caduceus-Ph 131k	0.995	0.626	256
Caduceus-Ps 131k	0.997	0.639	512
Evo 1	0.996	0.648	4096
Evo 2	0.995	0.643	4096

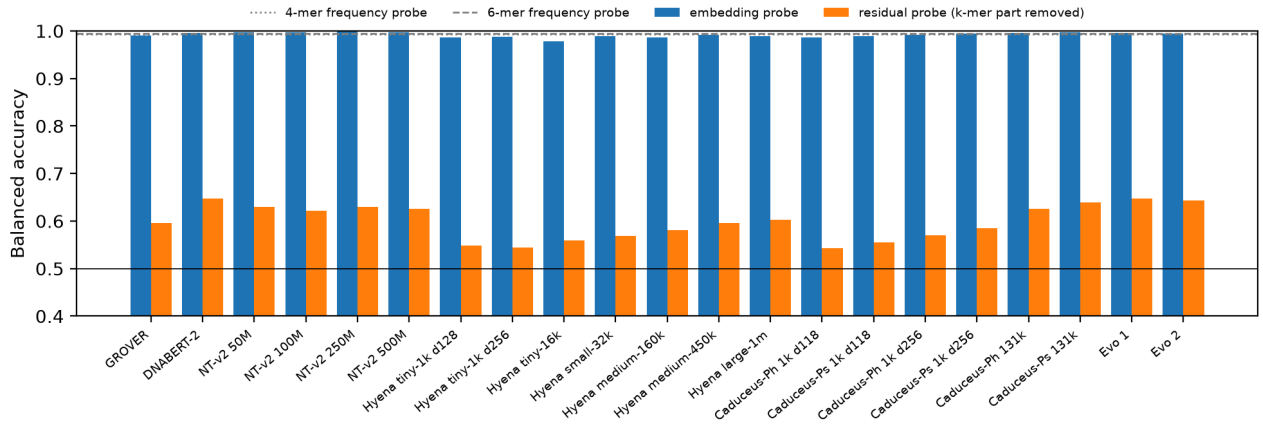


Figure 5: Embedding and residual probe accuracy per model, full length, CDS vs intron. Dotted and dashed lines: 4-mer and 6-mer probes.

5 Experiment B: perturbation ladder

5.1 Rationale

Experiment A shows *how much* is composition, but not *which* structure the models respond to. Experiment B edits test sequences so that each edit keeps some properties and destroys others. It then measures whether the unchanged probe still calls them coding. The edits form a ladder, from keeping only base composition to keeping the encoded protein. Comparing the steps shows which level of structure (composition, dinucleotides, codon usage, codon order, reading frame, strand) the coding score depends on.

5.2 Edits

All edits keep the sequence length. The reading frame is 0 for full-length CDS and the inferred frame for 200 bp coding windows. Non-coding sequences are edited as if read in frame 0, as a control.

Edit	Keeps	Destroys
original	everything	nothing (baseline)
nucleotide shuffle	base composition (GC)	all order
dinucleotide Markov	approx. dinucleotide frequencies: re-sampled from a first-order Markov chain fitted to the sequence	longer-range order, codons
codon shuffle	the multiset of codons (codon usage) and the frame	codon order, so the protein sequence
synonymous swap	the encoded protein: each codon is replaced by a random synonymous codon	codon usage and the original nucleotide sequence
stops in frame	everything except a few codon slots, which are overwritten with random stop codons in frame	the open reading frame
stops out of frame (control)	the same number of stop codons written at frame offset +1	same nucleotide damage as the in-frame edit, but the reading frame stays open
reverse complement	composition up to strand symmetry (GC is unchanged)	the strand and direction of the coding sequence

Stop-codon density. The number of inserted stops is $\max(3, \mathrm{round}(3L/200))$, which is three stops per 200 bp in both datasets. The out-of-frame control changes the same number of bases, so any difference between the two stop edits isolates the reading frame from the damage itself.

Sampling. Up to 2,000 test sequences per group, sampled with seed 0:

Data	Coding group	Non-coding group
200 bp	2,000 of the 8,824 windows with an inferred frame	2,000 of the 12,500 intergenomic windows
Full length	all 1,835 coding test sequences	2,000 intergenic / 2,000 lncRNA / all 1,808 intron test sequences

Edited sequences are re-embedded with the same model, chunking and pooling, and scored with the dataset's probe. The reported value is the fraction called coding (coding score > 0). In the full-length data the edited coding sequences are embedded once and scored by each of the three probes.

5.3 Results

In the tables: *nt shuffle* = nucleotide shuffle, *dinuc Markov* = dinucleotide Markov, *syn swap* = synonymous swap, *stops in/out* = stops in frame / out of frame, *rev comp* = reverse complement.

Table 11: Fraction of edited **coding** sequences called coding, 200 bp Genomic Benchmarks.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.932	0.929	0.773	0.947	0.985	0.868	0.880	0.779
DNABERT-2	0.933	0.906	0.737	0.949	0.988	0.866	0.893	0.787
NT-v2 50M	0.894	0.809	0.618	0.900	0.960	0.785	0.862	0.778
NT-v2 100M	0.897	0.837	0.631	0.916	0.965	0.773	0.850	0.735
NT-v2 250M	0.928	0.766	0.574	0.940	0.965	0.731	0.897	0.800
NT-v2 500M	0.903	0.842	0.661	0.925	0.961	0.798	0.856	0.779
Hyena tiny-1k d128	0.859	0.904	0.764	0.912	0.968	0.823	0.825	0.708
Hyena tiny-1k d256	0.884	0.904	0.751	0.918	0.966	0.837	0.839	0.727
Hyena tiny-16k	0.870	0.892	0.799	0.895	0.953	0.826	0.840	0.724
Hyena small-32k	0.889	0.914	0.779	0.923	0.973	0.856	0.868	0.747
Hyena medium-160k	0.898	0.873	0.747	0.901	0.962	0.836	0.868	0.712
Hyena medium-450k	0.858	0.885	0.746	0.900	0.963	0.822	0.827	0.719
Hyena large-1m	0.882	0.873	0.706	0.902	0.968	0.811	0.831	0.686
Caduceus-Ph 1k d118	0.875	0.923	0.769	0.912	0.977	0.843	0.863	0.741
Caduceus-Ps 1k d118	0.899	0.939	0.794	0.939	0.988	0.865	0.881	0.750
Caduceus-Ph 1k d256	0.912	0.934	0.735	0.943	0.985	0.873	0.896	0.757
Caduceus-Ps 1k d256	0.934	0.932	0.750	0.958	0.991	0.878	0.900	0.772
Caduceus-Ph 131k	0.943	0.972	0.808	0.970	0.996	0.906	0.928	0.858
Caduceus-Ps 131k	0.955	0.943	0.739	0.969	0.996	0.875	0.942	0.782
Evo 1	0.926	0.457	0.456	0.873	0.904	0.530	0.901	0.727
Evo 2	0.935	0.620	0.504	0.679	0.922	0.677	0.878	0.902

Table 12: Fraction of edited **non-coding** sequences (frame 0) called coding, 200 bp Genomic Benchmarks.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.100	0.633	0.292	0.441	0.733	0.076	0.076	0.091
DNABERT-2	0.090	0.555	0.258	0.417	0.771	0.069	0.074	0.093
NT-v2 50M	0.099	0.408	0.161	0.256	0.610	0.074	0.087	0.107
NT-v2 100M	0.095	0.452	0.173	0.284	0.621	0.075	0.081	0.113
NT-v2 250M	0.079	0.357	0.143	0.221	0.548	0.073	0.058	0.089
NT-v2 500M	0.102	0.439	0.191	0.288	0.630	0.078	0.083	0.116
Hyena tiny-1k d128	0.130	0.496	0.259	0.345	0.659	0.114	0.113	0.163
Hyena tiny-1k d256	0.133	0.516	0.257	0.355	0.682	0.107	0.106	0.142
Hyena tiny-16k	0.145	0.448	0.281	0.339	0.625	0.125	0.120	0.159
Hyena small-32k	0.124	0.498	0.283	0.378	0.673	0.119	0.118	0.139
Hyena medium-160k	0.118	0.433	0.257	0.334	0.590	0.104	0.102	0.134
Hyena medium-450k	0.136	0.444	0.226	0.312	0.637	0.116	0.113	0.151
Hyena large-1m	0.131	0.460	0.205	0.307	0.631	0.099	0.097	0.135
Caduceus-Ph 1k d118	0.131	0.509	0.280	0.394	0.642	0.120	0.117	0.145
Caduceus-Ps 1k d118	0.120	0.571	0.297	0.417	0.650	0.114	0.116	0.129
Caduceus-Ph 1k d256	0.098	0.557	0.242	0.385	0.649	0.088	0.092	0.123
Caduceus-Ps 1k d256	0.087	0.576	0.270	0.401	0.661	0.080	0.077	0.101
Caduceus-Ph 131k	0.075	0.671	0.330	0.491	0.678	0.074	0.072	0.070
Caduceus-Ps 131k	0.056	0.606	0.272	0.412	0.606	0.049	0.050	0.056
Evo 1	0.089	0.116	0.101	0.102	0.188	0.075	0.074	0.094
Evo 2	0.042	0.195	0.105	0.136	0.243	0.037	0.037	0.039



Figure 6: Perturbation ladder, 200 bp Genomic Benchmarks.

Table 13: Fraction of edited **coding** sequences called coding, full length, CDS vs intergenic.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.988	0.933	0.894	0.984	0.994	0.965	0.972	0.894
DNABERT-2	0.993	0.817	0.828	0.983	0.982	0.961	0.987	0.965
NT-v2 50M	0.998	0.258	0.243	0.989	0.990	0.831	0.997	0.719
NT-v2 100M	0.996	0.177	0.195	0.987	0.993	0.744	0.996	0.573
NT-v2 250M	0.997	0.076	0.123	0.991	0.985	0.634	0.995	0.462
NT-v2 500M	0.995	0.626	0.439	0.993	0.996	0.871	0.991	0.643
Hyena tiny-1k d128	0.983	0.962	0.907	0.972	0.996	0.952	0.963	0.778
Hyena tiny-1k d256	0.980	0.952	0.877	0.978	0.997	0.957	0.961	0.813
Hyena tiny-16k	0.978	0.965	0.922	0.978	0.997	0.951	0.960	0.803
Hyena small-32k	0.986	0.977	0.904	0.979	0.995	0.971	0.981	0.856
Hyena medium-160k	0.978	0.888	0.891	0.962	0.986	0.957	0.971	0.867
Hyena medium-450k	0.987	0.983	0.913	0.983	0.998	0.966	0.980	0.878
Hyena large-1m	0.983	0.959	0.909	0.969	0.996	0.959	0.971	0.899
Caduceus-Ph 1k d118	0.973	0.969	0.929	0.980	0.993	0.955	0.969	0.893
Caduceus-Ps 1k d118	0.982	0.936	0.943	0.979	0.987	0.962	0.973	0.838
Caduceus-Ph 1k d256	0.990	0.663	0.901	0.981	0.941	0.972	0.985	0.924
Caduceus-Ps 1k d256	0.995	0.968	0.916	0.991	0.994	0.985	0.990	0.923
Caduceus-Ph 131k	0.991	0.965	0.926	0.988	0.997	0.985	0.989	0.986
Caduceus-Ps 131k	0.996	0.975	0.858	0.997	0.999	0.967	0.993	0.850
Evo 1	0.992	0.192	0.192	0.934	0.991	0.318	0.991	0.900
Evo 2	0.992	0.118	0.171	0.265	0.947	0.705	0.975	0.958

Table 14: Fraction of edited **non-coding** sequences (frame 0) called coding, full length, CDS vs intergenic.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.012	0.570	0.198	0.339	0.622	0.005	0.004	0.013
DNABERT-2	0.006	0.398	0.167	0.399	0.703	0.003	0.003	0.005
NT-v2 50M	0.004	0.026	0.011	0.013	0.046	0.000	0.002	0.005
NT-v2 100M	0.005	0.025	0.006	0.013	0.064	0.001	0.006	0.005
NT-v2 250M	0.002	0.007	0.003	0.004	0.021	0.000	0.002	0.001
NT-v2 500M	0.003	0.199	0.025	0.064	0.304	0.000	0.004	0.007
Hyena tiny-1k d128	0.024	0.449	0.139	0.231	0.724	0.019	0.021	0.056
Hyena tiny-1k d256	0.022	0.460	0.140	0.231	0.701	0.017	0.014	0.045
Hyena tiny-16k	0.026	0.439	0.177	0.244	0.709	0.021	0.019	0.050
Hyena small-32k	0.013	0.598	0.197	0.333	0.719	0.014	0.012	0.023
Hyena medium-160k	0.018	0.297	0.186	0.205	0.537	0.015	0.015	0.038
Hyena medium-450k	0.011	0.614	0.225	0.318	0.770	0.011	0.011	0.038
Hyena large-1m	0.017	0.493	0.238	0.335	0.719	0.013	0.013	0.020
Caduceus-Ph 1k d118	0.028	0.468	0.214	0.324	0.572	0.028	0.029	0.036
Caduceus-Ps 1k d118	0.019	0.318	0.267	0.291	0.353	0.018	0.021	0.036
Caduceus-Ph 1k d256	0.013	0.415	0.237	0.395	0.281	0.013	0.011	0.019
Caduceus-Ps 1k d256	0.007	0.677	0.234	0.414	0.604	0.005	0.009	0.011
Caduceus-Ph 131k	0.005	0.712	0.310	0.516	0.559	0.005	0.005	0.004
Caduceus-Ps 131k	0.003	0.583	0.175	0.344	0.498	0.003	0.003	0.004
Evo 1	0.009	0.017	0.006	0.009	0.013	0.006	0.007	0.013
Evo 2	0.009	0.030	0.028	0.028	0.018	0.003	0.004	0.009

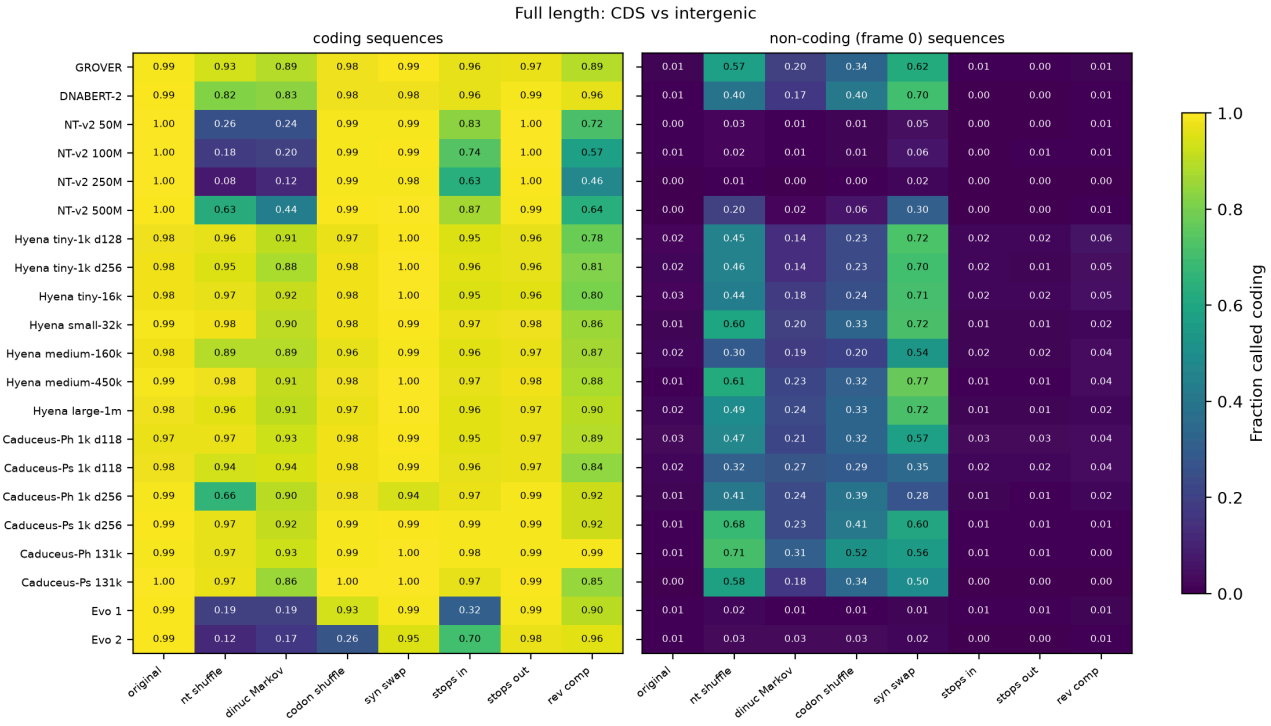


Figure 7: Perturbation ladder, full length, CDS vs intergenic.

Table 15: Fraction of edited **coding** sequences called coding, full length, CDS vs lncRNA.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.951	0.431	0.644	0.872	0.798	0.845	0.876	0.538
DNABERT-2	0.965	0.379	0.422	0.882	0.936	0.641	0.920	0.261
NT-v2 50M	0.993	0.074	0.047	0.969	0.947	0.492	0.980	0.354
NT-v2 100M	0.990	0.033	0.036	0.980	0.966	0.312	0.987	0.299
NT-v2 250M	0.991	0.017	0.033	0.960	0.967	0.202	0.993	0.068
NT-v2 500M	0.990	0.170	0.093	0.960	0.935	0.537	0.985	0.156
Hyena tiny-1k d128	0.933	0.884	0.736	0.935	0.957	0.856	0.878	0.696
Hyena tiny-1k d256	0.943	0.769	0.668	0.923	0.943	0.859	0.891	0.686
Hyena tiny-16k	0.934	0.833	0.808	0.936	0.934	0.857	0.893	0.681
Hyena small-32k	0.960	0.791	0.777	0.930	0.929	0.911	0.933	0.768
Hyena medium-160k	0.953	0.666	0.765	0.923	0.936	0.906	0.936	0.792
Hyena medium-450k	0.943	0.841	0.795	0.942	0.936	0.892	0.913	0.758
Hyena large-1m	0.951	0.810	0.762	0.911	0.963	0.892	0.916	0.757
Caduceus-Ph 1k d118	0.943	0.368	0.758	0.893	0.626	0.884	0.924	0.852
Caduceus-Ps 1k d118	0.961	0.642	0.849	0.951	0.829	0.908	0.944	0.796
Caduceus-Ph 1k d256	0.965	0.336	0.695	0.915	0.696	0.917	0.947	0.858
Caduceus-Ps 1k d256	0.975	0.474	0.749	0.946	0.868	0.930	0.954	0.769
Caduceus-Ph 131k	0.974	0.639	0.691	0.959	0.911	0.935	0.965	0.571
Caduceus-Ps 131k	0.986	0.099	0.416	0.958	0.844	0.848	0.981	0.135
Evo 1	0.986	0.005	0.020	0.777	0.946	0.081	0.984	0.857
Evo 2	0.979	0.007	0.005	0.011	0.743	0.057	0.936	0.922

Table 16: Fraction of edited **non-coding** sequences (frame 0) called coding, full length, CDS vs lncRNA.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.046	0.374	0.329	0.374	0.180	0.029	0.030	0.050
DNABERT-2	0.031	0.293	0.200	0.297	0.351	0.021	0.021	0.018
NT-v2 50M	0.009	0.024	0.018	0.019	0.013	0.001	0.007	0.013
NT-v2 100M	0.009	0.018	0.019	0.020	0.018	0.000	0.011	0.005
NT-v2 250M	0.005	0.003	0.004	0.007	0.005	0.000	0.005	0.004
NT-v2 500M	0.007	0.087	0.035	0.040	0.035	0.000	0.013	0.011
Hyena tiny-1k d128	0.081	0.782	0.322	0.534	0.635	0.059	0.060	0.091
Hyena tiny-1k d256	0.070	0.679	0.301	0.462	0.474	0.050	0.047	0.080
Hyena tiny-16k	0.093	0.708	0.398	0.534	0.466	0.062	0.067	0.102
Hyena small-32k	0.050	0.725	0.452	0.601	0.414	0.046	0.051	0.074
Hyena medium-160k	0.059	0.563	0.409	0.478	0.346	0.058	0.057	0.092
Hyena medium-450k	0.061	0.722	0.418	0.525	0.410	0.053	0.050	0.075
Hyena large-1m	0.062	0.671	0.388	0.500	0.461	0.046	0.052	0.079
Caduceus-Ph 1k d118	0.071	0.255	0.384	0.452	0.150	0.060	0.060	0.086
Caduceus-Ps 1k d118	0.066	0.406	0.514	0.579	0.232	0.056	0.056	0.071
Caduceus-Ph 1k d256	0.050	0.309	0.453	0.578	0.141	0.043	0.043	0.050
Caduceus-Ps 1k d256	0.038	0.441	0.423	0.501	0.203	0.028	0.030	0.047
Caduceus-Ph 131k	0.035	0.650	0.560	0.618	0.178	0.029	0.033	0.041
Caduceus-Ps 131k	0.021	0.117	0.313	0.286	0.078	0.018	0.017	0.029
Evo 1	0.013	0.004	0.003	0.006	0.007	0.007	0.009	0.010
Evo 2	0.011	0.000	0.002	0.000	0.001	0.004	0.004	0.009

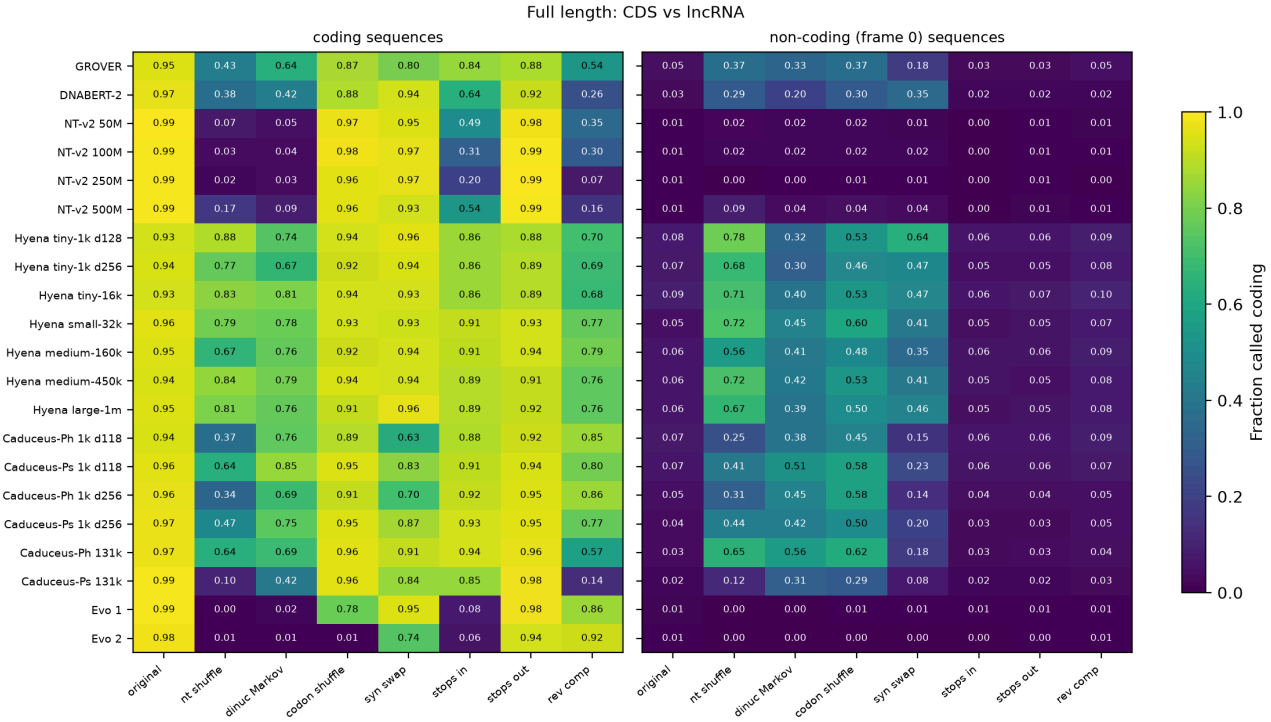


Figure 8: Perturbation ladder, full length, CDS vs lncRNA.

Table 17: Fraction of edited **coding** sequences called coding, full length, CDS vs intron.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.990	0.831	0.924	0.981	0.966	0.977	0.980	0.892
DNABERT-2	0.996	0.748	0.845	0.982	0.985	0.959	0.984	0.962
NT-v2 50M	0.998	0.066	0.283	0.987	0.943	0.875	0.990	0.826
NT-v2 100M	0.997	0.360	0.419	0.990	0.977	0.904	0.995	0.798
NT-v2 250M	0.998	0.072	0.478	0.990	0.973	0.840	0.997	0.584
NT-v2 500M	0.997	0.511	0.609	0.988	0.978	0.926	0.993	0.887
Hyena tiny-1k d128	0.989	0.932	0.922	0.982	0.978	0.969	0.975	0.802
Hyena tiny-1k d256	0.988	0.861	0.916	0.977	0.978	0.966	0.973	0.851
Hyena tiny-16k	0.983	0.904	0.935	0.981	0.974	0.965	0.969	0.788
Hyena small-32k	0.986	0.931	0.924	0.983	0.975	0.977	0.982	0.882
Hyena medium-160k	0.985	0.877	0.912	0.976	0.971	0.974	0.983	0.892
Hyena medium-450k	0.990	0.953	0.934	0.984	0.985	0.971	0.975	0.903
Hyena large-1m	0.990	0.942	0.914	0.973	0.988	0.964	0.975	0.847
Caduceus-Ph 1k d118	0.986	0.837	0.923	0.981	0.928	0.968	0.979	0.940
Caduceus-Ps 1k d118	0.990	0.821	0.933	0.985	0.960	0.975	0.984	0.886
Caduceus-Ph 1k d256	0.993	0.734	0.908	0.984	0.933	0.984	0.986	0.934
Caduceus-Ps 1k d256	0.996	0.700	0.902	0.980	0.927	0.991	0.991	0.929
Caduceus-Ph 131k	0.993	0.864	0.895	0.989	0.989	0.985	0.991	0.983
Caduceus-Ps 131k	0.995	0.663	0.791	0.990	0.977	0.976	0.995	0.967
Evo 1	0.993	0.066	0.082	0.922	0.979	0.228	0.991	0.969
Evo 2	0.991	0.009	0.013	0.014	0.894	0.678	0.974	0.931

Table 18: Fraction of edited **non-coding** sequences (frame 0) called coding, full length, CDS vs intron.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	0.010	0.426	0.253	0.280	0.304	0.004	0.004	0.055
DNABERT-2	0.003	0.434	0.144	0.299	0.568	0.001	0.001	0.019
NT-v2 50M	0.003	0.005	0.014	0.011	0.008	0.001	0.001	0.015
NT-v2 100M	0.003	0.160	0.015	0.023	0.148	0.000	0.001	0.012
NT-v2 250M	0.001	0.006	0.020	0.013	0.006	0.000	0.000	0.010
NT-v2 500M	0.003	0.173	0.066	0.056	0.159	0.001	0.002	0.018
Hyena tiny-1k d128	0.015	0.572	0.213	0.287	0.631	0.009	0.011	0.068
Hyena tiny-1k d256	0.012	0.444	0.168	0.215	0.352	0.006	0.006	0.066
Hyena tiny-16k	0.026	0.431	0.211	0.247	0.413	0.020	0.017	0.080
Hyena small-32k	0.008	0.603	0.235	0.340	0.409	0.011	0.009	0.062
Hyena medium-160k	0.012	0.511	0.185	0.252	0.418	0.013	0.011	0.064
Hyena medium-450k	0.007	0.688	0.232	0.332	0.617	0.006	0.004	0.064
Hyena large-1m	0.012	0.633	0.182	0.278	0.602	0.010	0.006	0.077
Caduceus-Ph 1k d118	0.014	0.356	0.239	0.288	0.248	0.013	0.013	0.082
Caduceus-Ps 1k d118	0.011	0.267	0.236	0.279	0.227	0.009	0.009	0.068
Caduceus-Ph 1k d256	0.009	0.282	0.257	0.305	0.171	0.008	0.007	0.051
Caduceus-Ps 1k d256	0.007	0.334	0.220	0.257	0.274	0.004	0.004	0.030
Caduceus-Ph 131k	0.002	0.666	0.319	0.482	0.441	0.005	0.003	0.019
Caduceus-Ps 131k	0.001	0.488	0.275	0.360	0.283	0.004	0.001	0.022
Evo 1	0.001	0.011	0.004	0.005	0.008	0.002	0.002	0.004
Evo 2	0.001	0.001	0.001	0.001	0.000	0.000	0.000	0.004

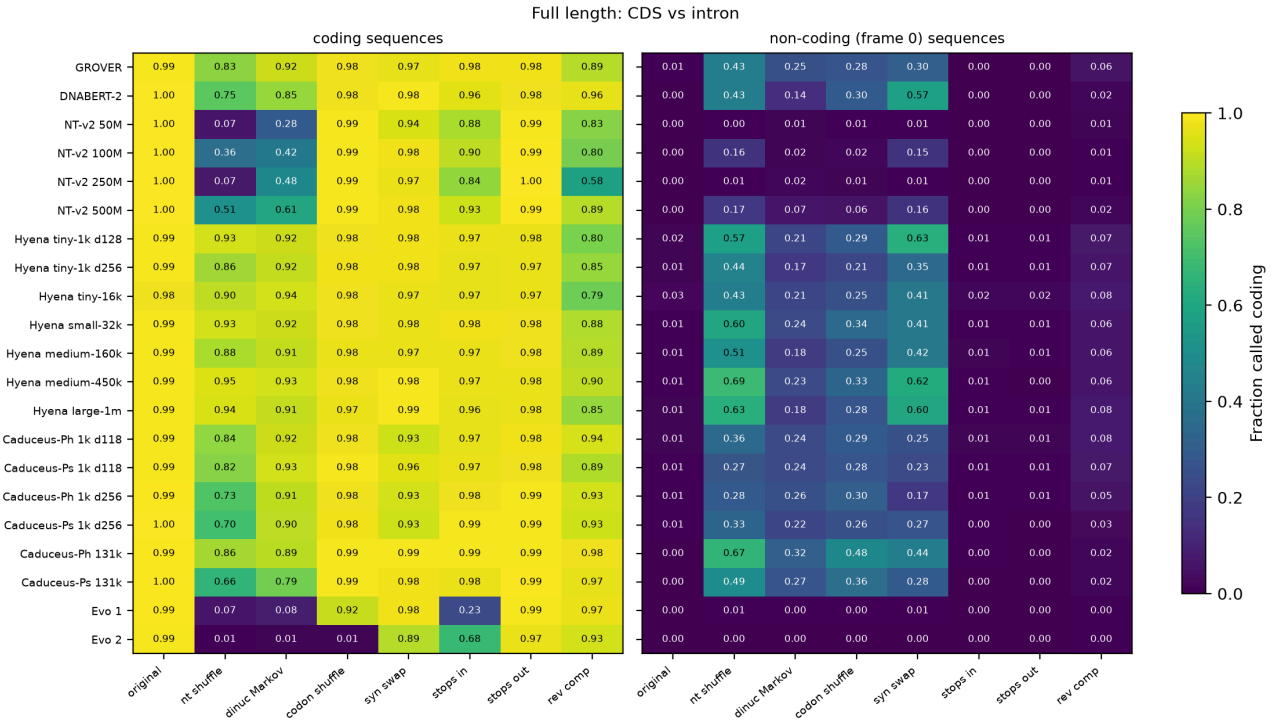


Figure 9: Perturbation ladder, full length, CDS vs intron.

Mean coding scores for every edit are in the appendix (Appendix A.1).

6 Experiment C: in-silico mutagenesis

6.1 Rationale

If a model represents codons, single-base mutations should affect the coding score depending on where they fall in the codon and what they do to the encoded protein:

- Under the genetic code, most changes at codon position 3 (the wobble position) are synonymous.
- Changes at positions 1 and 2 usually alter the amino acid.
- A nonsense mutation (a new stop codon) interrupts the reading frame.

Experiment C measures, base by base, how the coding score responds to every possible substitution. Non-coding sequences mutated the same way (read in frame 0) give the control, where codon position and mutation class have no biological meaning.

6.2 Method

- For each selected sequence and each selected codon, every base of the codon is changed to each of the three other bases (9 variants per codon).
- Each variant is embedded and scored, and the effect is Δ = coding score(variant) − coding score(original).
- Each variant is labelled with its codon position (1, 2 or 3) and its mutation class from the standard genetic code:
  - *synonymous*: same amino acid;
  - *missense*: different amino acid;
  - *nonsense*: sense codon to stop codon;
  - *stop-loss* and *stop-to-stop*: these occur only when the mutated codon was already a stop codon.

	200 bp	**Full length**
Coding sequences	300 windows with an inferred frame	100 coding test sequences
Control sequences	300 intergenomic windows (frame 0)	100 intergenic test sequences (frame 0)
Codons mutated	every complete codon (66 per window, 594 variants)	60 random codons per sequence (540 variants)
Probes	the 200 bp probe	each of the three full-length probes scores the same variants

In the full-length data the same 100 coding and 100 intergenic sequences, and the same variants, are used for all three datasets; only the probe changes. The intergenic control therefore also serves as the control for the lncRNA and intron datasets. Because the final codon of each CDS is its stop codon, a small number of full-length coding variants are stop-loss or stop-to-stop; these are left out of the tables below.

Summary statistics.

- *Codon position*: mean $|\Delta|$ at each codon position, and the ratio of position 3 to the mean of positions 1 and 2. A ratio below 1 means wobble-position mutations change the score less.
- *Mutation class*: mean signed Δ per class. A negative value means mutations of that class move sequences towards non-coding.

6.3 Results

Table 19: Mean |Δ| coding score by codon position, and the position-3 / position-1-2 ratio, 200 bp Genomic Benchmarks.

Model	cod. pos1	cod. pos2	cod. pos3	ctrl pos1	ctrl pos2	ctrl pos3	cod. ratio	ctrl ratio
GROVER	0.326	0.319	0.329	0.310	0.309	0.306	1.022	0.991
DNABERT-2	0.384	0.374	0.380	0.468	0.467	0.471	1.003	1.008
NT-v2 50M	0.362	0.342	0.353	0.351	0.356	0.351	1.002	0.992
NT-v2 100M	0.350	0.326	0.333	0.338	0.349	0.346	0.986	1.009
NT-v2 250M	0.435	0.379	0.380	0.357	0.358	0.351	0.932	0.982
NT-v2 500M	0.393	0.374	0.378	0.422	0.426	0.422	0.986	0.996
Hyena tiny-1k d128	0.195	0.191	0.197	0.241	0.241	0.239	1.024	0.992
Hyena tiny-1k d256	0.229	0.225	0.230	0.266	0.265	0.264	1.012	0.996
Hyena tiny-16k	0.185	0.181	0.184	0.270	0.269	0.270	1.006	1.000
Hyena small-32k	0.225	0.216	0.222	0.292	0.290	0.291	1.007	1.001
Hyena medium-160k	0.232	0.224	0.233	0.240	0.239	0.239	1.022	0.997
Hyena medium-450k	0.223	0.215	0.224	0.213	0.212	0.211	1.022	0.996
Hyena large-1m	0.217	0.208	0.218	0.217	0.217	0.218	1.029	1.004
Caduceus-Ph 1k d118	0.189	0.182	0.190	0.253	0.255	0.257	1.022	1.008
Caduceus-Ps 1k d118	0.198	0.192	0.197	0.316	0.317	0.314	1.011	0.992
Caduceus-Ph 1k d256	0.246	0.235	0.262	0.274	0.273	0.271	1.090	0.993
Caduceus-Ps 1k d256	0.270	0.259	0.288	0.278	0.278	0.278	1.091	1.000
Caduceus-Ph 131k	0.377	0.350	0.366	0.217	0.217	0.217	1.006	0.999
Caduceus-Ps 131k	0.432	0.403	0.355	0.238	0.235	0.237	0.851	1.004
Evo 1	0.588	0.539	0.509	0.489	0.490	0.487	0.904	0.995
Evo 2	1.807	2.010	1.120	0.249	0.249	0.249	0.587	1.002

Table 20: Mean signed Δ coding score by mutation class, 200 bp Genomic Benchmarks.

Model	cod. syn	cod. mis	cod. non	ctrl syn	ctrl mis	ctrl non
GROVER	0.026	0.015	-0.268	0.118	0.090	-0.063
DNABERT-2	0.022	0.012	-0.393	0.150	0.122	-0.078
NT-v2 50M	-0.000	-0.004	-0.507	0.120	0.088	-0.093
NT-v2 100M	0.019	0.012	-0.537	0.120	0.084	-0.117
NT-v2 250M	-0.007	-0.016	-1.202	0.132	0.095	-0.144
NT-v2 500M	0.018	0.016	-0.538	0.148	0.110	-0.127
Hyena tiny-1k d128	0.024	0.018	-0.123	0.096	0.081	-0.019
Hyena tiny-1k d256	0.030	0.020	-0.160	0.106	0.084	-0.036
Hyena tiny-16k	0.015	0.013	-0.097	0.110	0.090	-0.004
Hyena small-32k	0.021	0.018	-0.111	0.124	0.118	0.030
Hyena medium-160k	0.016	0.012	-0.164	0.084	0.075	-0.013
Hyena medium-450k	0.023	0.020	-0.124	0.082	0.062	-0.026
Hyena large-1m	0.017	0.016	-0.176	0.086	0.064	-0.063
Caduceus-Ph 1k d118	0.031	0.021	-0.091	0.095	0.085	0.022
Caduceus-Ps 1k d118	0.034	0.024	-0.113	0.111	0.092	-0.014
Caduceus-Ph 1k d256	0.005	0.009	-0.172	0.098	0.087	-0.019
Caduceus-Ps 1k d256	0.013	0.015	-0.230	0.109	0.087	-0.068
Caduceus-Ph 131k	0.029	0.013	-0.307	0.094	0.075	-0.020
Caduceus-Ps 131k	0.007	-0.017	-0.783	0.100	0.083	0.003
Evo 1	-0.052	-0.084	-1.632	0.035	0.024	-0.135
Evo 2	-0.085	-1.157	-5.118	0.077	0.066	0.025

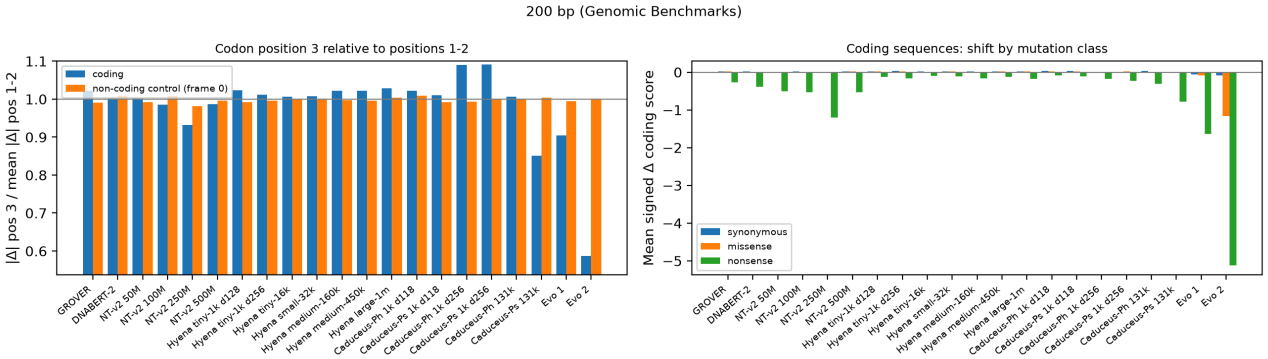


Figure 10: Codon-position ratio (coding vs control) and the signed shift by mutation class for coding sequences, 200 bp Genomic Benchmarks.

Table 21: Mean $|\Delta|$ coding score by codon position, and the position-3 / position-1-2 ratio, full length, CDS vs intergenic.

Model	cod. pos1	cod. pos2	cod. pos3	ctrl pos1	ctrl pos2	ctrl pos3	cod. ratio	ctrl ratio
GROVER	0.433	0.424	0.433	0.318	0.317	0.314	1.010	0.989
DNABERT-2	0.471	0.467	0.460	0.345	0.351	0.354	0.981	1.017
NT-v2 50M	0.400	0.362	0.352	0.316	0.306	0.300	0.924	0.964
NT-v2 100M	0.372	0.352	0.319	0.327	0.326	0.320	0.880	0.979
NT-v2 250M	0.472	0.421	0.375	0.329	0.321	0.319	0.840	0.983
NT-v2 500M	0.564	0.522	0.488	0.450	0.443	0.435	0.899	0.974
Hyena tiny-1k d128	0.129	0.131	0.133	0.150	0.149	0.151	1.025	1.011
Hyena tiny-1k d256	0.165	0.165	0.169	0.180	0.178	0.179	1.021	0.997
Hyena tiny-16k	0.102	0.104	0.104	0.153	0.152	0.152	1.014	0.993
Hyena small-32k	0.159	0.162	0.161	0.201	0.202	0.204	1.007	1.009
Hyena medium-160k	0.134	0.130	0.132	0.150	0.148	0.149	0.999	0.996
Hyena medium-450k	0.159	0.162	0.164	0.157	0.160	0.156	1.022	0.987
Hyena large-1m	0.125	0.123	0.126	0.142	0.143	0.142	1.014	0.996
Caduceus-Ph 1k d118	0.096	0.097	0.097	0.127	0.128	0.130	1.009	1.024
Caduceus-Ps 1k d118	0.113	0.110	0.112	0.150	0.149	0.151	1.007	1.009
Caduceus-Ph 1k d256	0.146	0.136	0.155	0.140	0.141	0.141	1.100	1.004
Caduceus-Ps 1k d256	0.141	0.137	0.151	0.152	0.151	0.148	1.085	0.979
Caduceus-Ph 131k	0.175	0.170	0.168	0.098	0.098	0.098	0.974	0.996
Caduceus-Ps 131k	0.173	0.158	0.160	0.106	0.107	0.106	0.967	0.990
Evo 1	0.431	0.379	0.370	0.624	0.625	0.625	0.913	1.001
Evo 2	0.701	0.715	0.413	0.071	0.070	0.070	0.584	0.989

Table 22: Mean signed Δ coding score by mutation class, full length, CDS vs intergenic.

Model	cod. syn	cod. mis	cod. non	ctrl syn	ctrl mis	ctrl non
GROVER	0.016	0.007	-0.298	0.070	0.053	-0.072
DNABERT-2	0.011	-0.009	-0.371	0.037	0.029	-0.047
NT-v2 50M	-0.018	-0.013	-0.976	0.063	0.057	-0.258
NT-v2 100M	0.015	0.018	-0.764	0.058	0.048	-0.210
NT-v2 250M	-0.010	-0.007	-1.727	0.086	0.063	-0.152
NT-v2 500M	-0.013	-0.018	-0.880	0.102	0.066	-0.221
Hyena tiny-1k d128	0.010	0.002	-0.084	0.046	0.043	-0.015
Hyena tiny-1k d256	0.012	0.005	-0.107	0.059	0.047	-0.014
Hyena tiny-16k	0.011	0.005	-0.056	0.042	0.044	-0.006
Hyena small-32k	0.010	0.001	-0.081	0.082	0.080	0.047
Hyena medium-160k	-0.003	-0.003	-0.082	0.045	0.046	0.019
Hyena medium-450k	0.007	0.005	-0.103	0.054	0.051	0.026
Hyena large-1m	0.010	0.006	-0.088	0.045	0.039	-0.006
Caduceus-Ph 1k d118	0.009	0.005	-0.033	0.040	0.037	0.003
Caduceus-Ps 1k d118	0.009	0.008	-0.081	0.036	0.046	0.015
Caduceus-Ph 1k d256	-0.017	-0.003	-0.111	0.032	0.031	0.019
Caduceus-Ps 1k d256	-0.003	0.004	-0.097	0.046	0.047	0.010
Caduceus-Ph 131k	0.005	0.002	-0.167	0.034	0.033	0.008
Caduceus-Ps 131k	-0.003	-0.005	-0.276	0.028	0.026	0.002
Evo 1	0.019	-0.004	-1.528	0.422	0.413	0.416
Evo 2	-0.013	-0.266	-2.810	0.011	0.007	-0.010

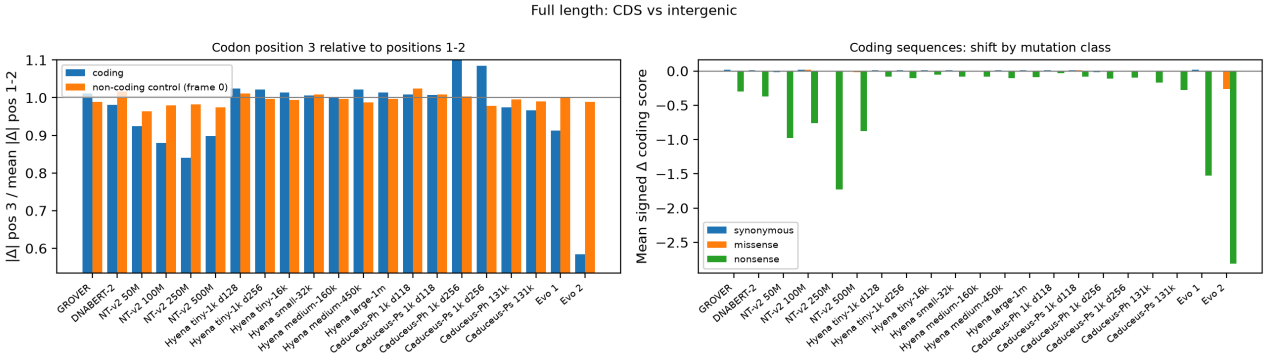


Figure 11: Codon-position ratio (coding vs control) and the signed shift by mutation class for coding sequences, full length, CDS vs intergenic.

Table 23: Mean $|\Delta|$ coding score by codon position, and the position-3 / position-1-2 ratio, full length, CDS vs lncRNA.

Model	cod. pos1	cod. pos2	cod. pos3	ctrl pos1	ctrl pos2	ctrl pos3	cod. ratio	ctrl ratio
GROVER	0.286	0.276	0.284	0.296	0.291	0.293	1.011	0.997
DNABERT-2	0.468	0.463	0.444	0.610	0.624	0.614	0.955	0.995
NT-v2 50M	0.330	0.310	0.303	0.289	0.283	0.283	0.947	0.990
NT-v2 100M	0.316	0.296	0.267	0.316	0.315	0.307	0.873	0.975
NT-v2 250M	0.398	0.362	0.327	0.345	0.337	0.337	0.859	0.988
NT-v2 500M	0.460	0.437	0.405	0.431	0.420	0.420	0.903	0.987
Hyena tiny-1k d128	0.118	0.119	0.120	0.160	0.159	0.161	1.011	1.011
Hyena tiny-1k d256	0.141	0.140	0.145	0.175	0.175	0.175	1.030	0.998
Hyena tiny-16k	0.106	0.108	0.108	0.181	0.181	0.179	1.011	0.989
Hyena small-32k	0.132	0.134	0.133	0.195	0.199	0.196	1.001	0.994
Hyena medium-160k	0.113	0.113	0.113	0.154	0.152	0.152	1.004	0.993
Hyena medium-450k	0.120	0.122	0.124	0.147	0.148	0.148	1.021	1.001
Hyena large-1m	0.103	0.101	0.102	0.135	0.135	0.134	1.001	0.990
Caduceus-Ph 1k d118	0.093	0.091	0.094	0.143	0.145	0.146	1.020	1.018
Caduceus-Ps 1k d118	0.088	0.088	0.090	0.169	0.165	0.170	1.024	1.016
Caduceus-Ph 1k d256	0.100	0.100	0.102	0.141	0.140	0.141	1.025	1.000
Caduceus-Ps 1k d256	0.108	0.107	0.115	0.143	0.141	0.141	1.063	0.995
Caduceus-Ph 131k	0.099	0.096	0.096	0.096	0.095	0.097	0.979	1.010
Caduceus-Ps 131k	0.124	0.116	0.117	0.110	0.109	0.109	0.980	0.994
Evo 1	0.563	0.526	0.522	1.133	1.137	1.131	0.958	0.996
Evo 2	0.298	0.248	0.168	0.060	0.059	0.060	0.615	1.008

Table 24: Mean signed Δ coding score by mutation class, full length, CDS vs lncRNA.

Model	cod. syn	cod. mis	cod. non	ctrl syn	ctrl mis	ctrl non
GROVER	-0.001	-0.002	-0.176	0.052	0.037	-0.043
DNABERT-2	-0.018	-0.026	-0.434	0.150	0.109	0.082
NT-v2 50M	-0.024	-0.023	-0.747	-0.001	0.004	-0.249
NT-v2 100M	-0.001	-0.002	-0.742	0.024	0.016	-0.208
NT-v2 250M	-0.016	-0.014	-1.376	0.038	0.038	-0.174
NT-v2 500M	-0.016	-0.008	-0.793	0.022	0.011	-0.282
Hyena tiny-1k d128	0.006	0.002	-0.064	0.046	0.044	-0.007
Hyena tiny-1k d256	0.006	0.002	-0.086	0.046	0.037	-0.034
Hyena tiny-16k	0.006	0.001	-0.056	0.043	0.051	-0.007
Hyena small-32k	0.007	0.002	-0.051	0.067	0.068	0.034
Hyena medium-160k	-0.003	-0.004	-0.060	0.043	0.042	0.027
Hyena medium-450k	0.003	-0.000	-0.079	0.046	0.043	0.014
Hyena large-1m	0.001	-0.001	-0.054	0.038	0.037	-0.002
Caduceus-Ph 1k d118	-0.002	-0.002	-0.057	0.029	0.025	-0.019
Caduceus-Ps 1k d118	0.003	0.004	-0.062	0.040	0.047	0.001
Caduceus-Ph 1k d256	0.003	-0.000	-0.070	0.037	0.035	-0.003
Caduceus-Ps 1k d256	-0.001	0.000	-0.090	0.042	0.039	-0.004
Caduceus-Ph 131k	-0.004	-0.001	-0.085	0.021	0.025	0.017
Caduceus-Ps 131k	-0.000	-0.001	-0.216	0.034	0.035	0.019
Evo 1	0.275	0.247	-1.144	0.958	0.948	0.957
Evo 2	-0.009	-0.077	-2.049	-0.009	-0.009	-0.011

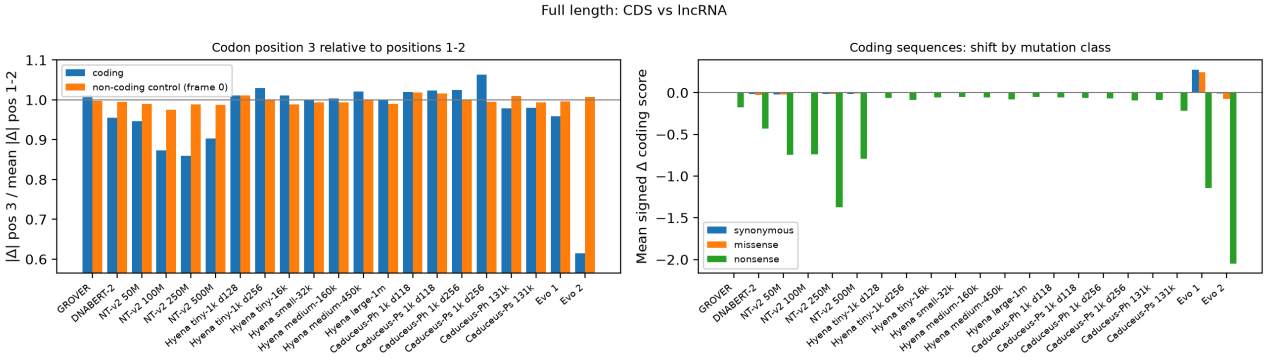


Figure 12: Codon-position ratio (coding vs control) and the signed shift by mutation class for coding sequences, full length, CDS vs lncRNA.

Table 25: Mean $|\Delta|$ coding score by codon position, and the position-3 / position-1-2 ratio, full length, CDS vs intron.

Model	cod. pos1	cod. pos2	cod. pos3	ctrl pos1	ctrl pos2	ctrl pos3	cod. ratio	ctrl ratio
GROVER	0.452	0.445	0.446	0.356	0.355	0.355	0.994	0.998
DNABERT-2	0.523	0.528	0.513	0.477	0.480	0.474	0.977	0.989
NT-v2 50M	0.414	0.393	0.364	0.298	0.299	0.295	0.902	0.987
NT-v2 100M	0.362	0.357	0.320	0.321	0.325	0.321	0.888	0.994
NT-v2 250M	0.456	0.422	0.375	0.314	0.309	0.304	0.854	0.976
NT-v2 500M	0.529	0.503	0.471	0.442	0.442	0.428	0.913	0.967
Hyena tiny-1k d128	0.162	0.161	0.163	0.185	0.186	0.188	1.007	1.013
Hyena tiny-1k d256	0.198	0.200	0.198	0.214	0.211	0.212	0.993	0.997
Hyena tiny-16k	0.125	0.128	0.127	0.178	0.177	0.182	1.004	1.026
Hyena small-32k	0.191	0.195	0.193	0.217	0.218	0.217	1.001	0.997
Hyena medium-160k	0.157	0.155	0.157	0.165	0.161	0.162	1.008	0.995
Hyena medium-450k	0.195	0.195	0.198	0.171	0.173	0.170	1.018	0.989
Hyena large-1m	0.149	0.148	0.150	0.153	0.155	0.152	1.010	0.990
Caduceus-Ph 1k d118	0.129	0.127	0.129	0.144	0.147	0.148	1.008	1.018
Caduceus-Ps 1k d118	0.121	0.122	0.122	0.160	0.157	0.162	1.002	1.024
Caduceus-Ph 1k d256	0.156	0.160	0.168	0.155	0.155	0.155	1.063	1.005
Caduceus-Ps 1k d256	0.171	0.170	0.185	0.150	0.152	0.149	1.089	0.985
Caduceus-Ph 131k	0.186	0.178	0.173	0.103	0.101	0.103	0.948	1.008
Caduceus-Ps 131k	0.200	0.196	0.200	0.113	0.111	0.111	1.010	0.992
Evo 1	0.875	0.833	0.812	0.980	0.983	0.979	0.950	0.997
Evo 2	0.645	0.671	0.386	0.066	0.066	0.065	0.587	0.987

Table 26: Mean signed Δ coding score by mutation class, full length, CDS vs intron.

Model	cod. syn	cod. mis	cod. non	ctrl syn	ctrl mis	ctrl non
GROVER	-0.011	-0.009	-0.350	0.066	0.046	-0.099
DNABERT-2	0.009	-0.017	-0.493	0.110	0.089	-0.044
NT-v2 50M	-0.034	-0.032	-0.947	0.026	0.025	-0.241
NT-v2 100M	0.010	0.002	-0.683	0.038	0.027	-0.184
NT-v2 250M	-0.014	-0.015	-1.325	0.049	0.038	-0.122
NT-v2 500M	-0.034	-0.031	-0.679	0.066	0.040	-0.193
Hyena tiny-1k d128	0.007	0.000	-0.112	0.050	0.046	-0.022
Hyena tiny-1k d256	0.004	-0.003	-0.156	0.047	0.041	-0.073
Hyena tiny-16k	0.001	-0.003	-0.091	0.027	0.036	-0.034
Hyena small-32k	0.001	-0.007	-0.110	0.051	0.052	0.004
Hyena medium-160k	-0.003	-0.008	-0.112	0.033	0.027	0.001
Hyena medium-450k	0.002	-0.001	-0.139	0.037	0.030	-0.014
Hyena large-1m	0.002	-0.004	-0.113	0.035	0.029	-0.024
Caduceus-Ph 1k d118	0.002	-0.003	-0.090	0.025	0.022	-0.028
Caduceus-Ps 1k d118	0.005	0.001	-0.099	0.031	0.040	-0.007
Caduceus-Ph 1k d256	-0.020	-0.006	-0.133	0.025	0.024	-0.004
Caduceus-Ps 1k d256	-0.024	-0.007	-0.143	0.034	0.028	-0.016
Caduceus-Ph 131k	0.002	-0.002	-0.172	0.026	0.033	0.012
Caduceus-Ps 131k	-0.010	-0.004	-0.250	0.020	0.021	0.006
Evo 1	0.469	0.414	-1.613	0.816	0.809	0.799
Evo 2	-0.009	-0.195	-2.250	0.001	-0.001	-0.006

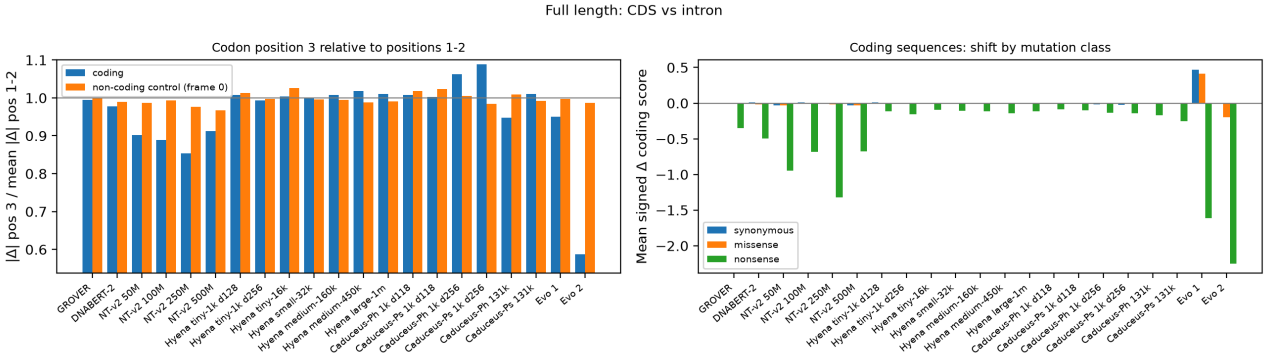


Figure 13: Codon-position ratio (coding vs control) and the signed shift by mutation class for coding sequences, full length, CDS vs intron.

7 Experiment D: non-coding signatures

7.1 Rationale

Experiments A–C ask what makes a sequence look coding. Non-coding DNA, however, is heterogeneous: in the human genome it is largely transposable elements (Alu, L1, MIR, L2, LTR elements, DNA transposons), simple repeats and low-complexity sequence, with distinct composition. Experiment D asks which properties of the *non-coding* test sequences go with being recognised as non-coding. Each non-coding sequence gets interpretable features, and these are related to its coding score under each model.

7.2 Features, 200 bp data (`noncoding_signatures.py`)

Repeat family by alignment.

- Each non-coding test window (12,500) is locally aligned on both strands to the Dfam consensus sequences of representative human repeat families:
  - Alu: AluY, AluSx, AluJb
  - MIR
  - L1: the 3′ ends of L1HS, L1PA7 and L1MB7
  - L2: the 3′ ends of L2a and L2b
  - LTR: THE1B, MLT1A0, MSTA, LTR12C
  - DNA transposons: MER5A, Tigger1
- The 3′ ends are used for LINEs because most genomic LINE copies are 5′-truncated.
- Alignment scoring: match +2, mismatch −3, gap open −5, gap extension −2.

Significance threshold. For each family, the threshold is the 99.5th percentile of best scores for 1,000 shuffled windows. A window is assigned the class of the family it beats by the largest margin. Windows with no significant hit are labelled *none*, or *simple repeat* if their tandem-repeat score (below) exceeds 0.6.

7.3 Features, full-length data (`noncoding_signatures_full.py`)

Repeat coverage.

- RepeatMasker coverage is taken from the UCSC hg38 `rmsk` table, using each sequence's genomic intervals (exons for spliced lncRNAs).
- For each repeat class, the feature is the fraction of the sequence's bases it covers. Classes: Alu, L1, MIR, L2, LTR, DNA transposon, simple repeat, low complexity, other repeat.
- The real annotation replaces consensus alignment because full-length sequences contain many repeats of every age.
- A sequence's *dominant class* is the class with the highest coverage if that coverage is at least 10%; otherwise it is *none*.

7.4 Sequence features (both data types)

Computed from the sequence and normalised for length:

- GC fraction.
- CpG observed/expected: $n_{\mathrm{CG}} \cdot L/(n_C \cdot n_G)$.
- Longest homopolymer run.
- Tandem-repeat score: the maximum over periods $p = 1..6$ of the fraction of positions equal to the base p positions later.
- Trinucleotide Shannon entropy (low means low complexity).

7.5 Statistics

For each model and dataset:

- The probe trained on that dataset scores the non-coding test sequences.
- The fraction called non-coding (coding score < 0) is reported per repeat class.
- The Spearman correlation of each feature with the coding score is computed.
- The cross-validated R^2 (5-fold, ordinary least squares) is computed for predicting the coding score from all features: the continuous features plus one-hot repeat class for 200 bp; the coverages plus continuous features for full length.

7.6 Results

Column headers of the repeat-class tables give the number of sequences in each class in parentheses. Abbreviations: *DNA* = DNA transposon, *simple* = simple repeat, *low cx* = low complexity, *other* = other repeat, *CpG o/e* = CpG observed/expected, *homopol.* = longest homopolymer, *tandem* = tandem-repeat score, *entropy* = trinucleotide entropy. In the full-length correlation tables the repeat-class columns are coverage fractions.

Table 27: Fraction of non-coding test sequences called non-coding, by best-matching repeat class, 200 bp Genomic Benchmarks.

model	none (8011)	Alu (2024)	L1 (994)	LTR (616)	MIR (360)	L2 (296)	DNA (152)	simple (47)
GROVER	0.867	0.974	0.973	0.907	0.953	0.932	0.829	0.979
DNABERT-2	0.879	0.978	0.972	0.911	0.944	0.912	0.901	0.979
NT-v2 50M	0.882	0.965	0.951	0.830	0.914	0.895	0.862	1.000
NT-v2 100M	0.876	0.959	0.937	0.831	0.917	0.899	0.829	0.851
NT-v2 250M	0.902	0.968	0.959	0.852	0.939	0.905	0.829	0.979
NT-v2 500M	0.877	0.961	0.953	0.828	0.919	0.905	0.829	1.000
Hyena tiny-1k d128	0.831	0.971	0.962	0.823	0.864	0.824	0.743	0.957
Hyena tiny-1k d256	0.841	0.974	0.957	0.859	0.881	0.865	0.743	0.979
Hyena tiny-16k	0.820	0.975	0.948	0.813	0.856	0.821	0.711	0.957
Hyena small-32k	0.834	0.979	0.963	0.880	0.933	0.875	0.809	0.936
Hyena medium-160k	0.843	0.971	0.967	0.877	0.894	0.865	0.737	1.000
Hyena medium-450k	0.843	0.963	0.952	0.828	0.861	0.868	0.783	0.979
Hyena large-1m	0.853	0.952	0.939	0.792	0.906	0.882	0.776	0.894
Caduceus-Ph 1k d118	0.833	0.973	0.964	0.847	0.922	0.885	0.776	0.915
Caduceus-Ps 1k d118	0.843	0.978	0.967	0.870	0.961	0.902	0.757	1.000
Caduceus-Ph 1k d256	0.869	0.980	0.978	0.899	0.975	0.929	0.783	0.936
Caduceus-Ps 1k d256	0.883	0.981	0.975	0.912	0.956	0.943	0.809	0.979
Caduceus-Ph 131k	0.894	0.982	0.980	0.937	0.978	0.970	0.928	1.000
Caduceus-Ps 131k	0.928	0.986	0.985	0.943	0.983	0.963	0.947	1.000
Evo 1	0.901	0.912	0.933	0.830	0.928	0.899	0.901	0.979
Evo 2	0.950	0.986	0.990	0.924	0.967	0.936	0.947	0.957

Table 28: Spearman correlation of each feature with the coding score of non-coding test sequences, and 5-fold CV R^2 of a linear model from all features to the coding score, 200 bp Genomic Benchmarks.

model	GC	CpG o/e	homopol.	tandem	entropy	CV R^2
GROVER	0.35	0.08	-0.20	-0.26	0.29	0.26
DNABERT-2	0.32	0.11	-0.19	-0.25	0.26	0.20
NT-v2 50M	0.44	0.20	-0.27	-0.34	0.31	0.40
NT-v2 100M	0.53	0.26	-0.29	-0.39	0.36	0.46
NT-v2 250M	0.48	0.23	-0.30	-0.41	0.35	0.45
NT-v2 500M	0.49	0.20	-0.31	-0.44	0.38	0.48
Hyena tiny-1k d128	0.39	0.11	-0.34	-0.38	0.35	0.44
Hyena tiny-1k d256	0.38	0.13	-0.32	-0.39	0.38	0.39
Hyena tiny-16k	0.39	0.08	-0.36	-0.39	0.35	0.41
Hyena small-32k	0.32	0.11	-0.26	-0.31	0.32	0.28
Hyena medium-160k	0.42	0.10	-0.31	-0.39	0.39	0.42
Hyena medium-450k	0.47	0.12	-0.35	-0.42	0.41	0.50
Hyena large-1m	0.48	0.12	-0.36	-0.44	0.41	0.48
Caduceus-Ph 1k d118	0.39	0.12	-0.29	-0.36	0.35	0.32
Caduceus-Ps 1k d118	0.37	0.12	-0.27	-0.34	0.35	0.34
Caduceus-Ph 1k d256	0.38	0.14	-0.25	-0.34	0.38	0.29
Caduceus-Ps 1k d256	0.36	0.13	-0.22	-0.32	0.38	0.35
Caduceus-Ph 131k	0.36	0.11	-0.16	-0.28	0.35	0.27
Caduceus-Ps 131k	0.31	0.06	-0.12	-0.21	0.30	0.21
Evo 1	0.69	0.17	-0.25	-0.41	0.41	0.50
Evo 2	0.32	0.12	-0.25	-0.20	0.24	0.29

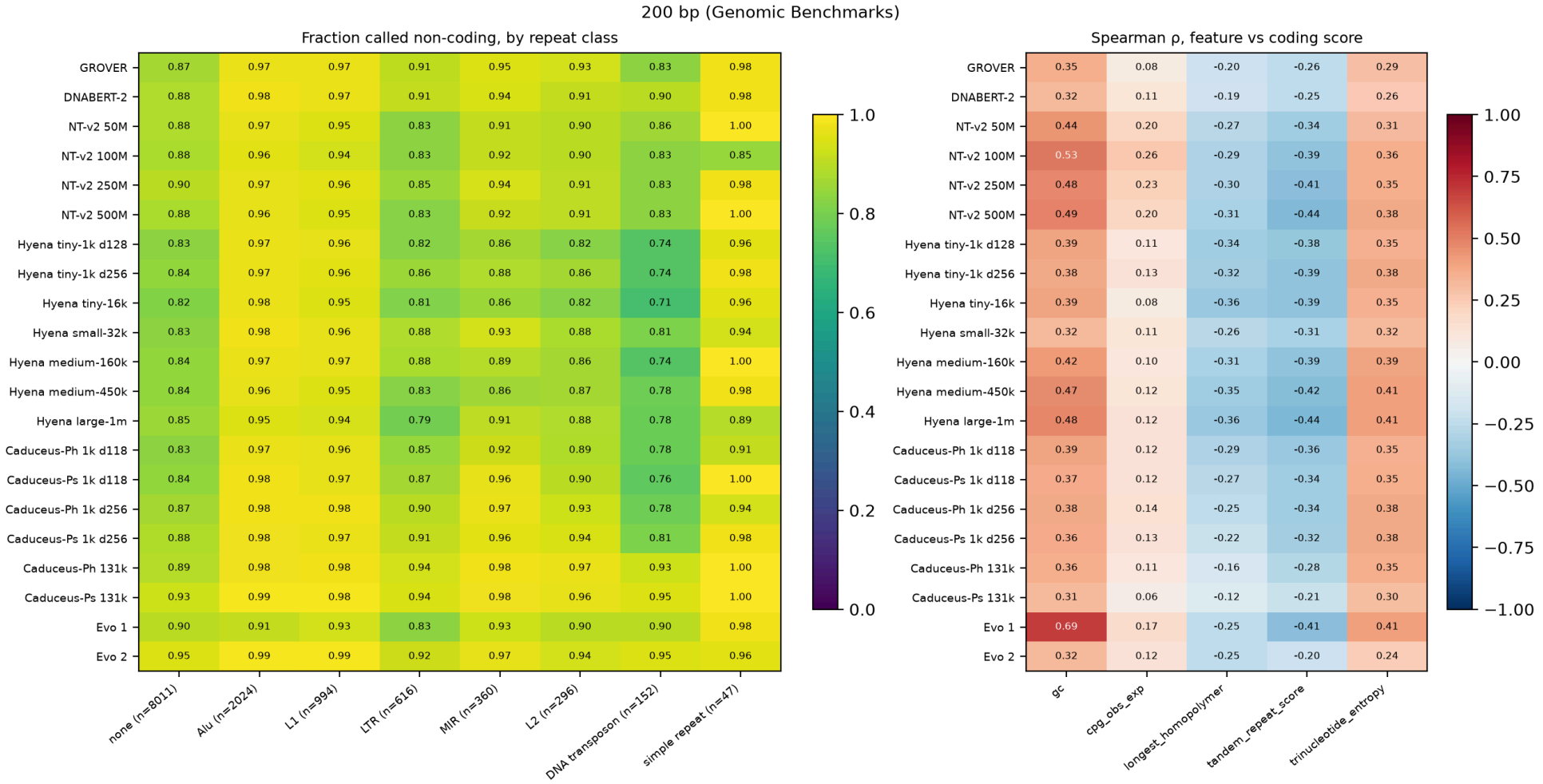


Figure 14: Fraction called non-coding by repeat class (left) and feature correlations with the coding score (right), 200 bp Genomic Benchmarks.

Table 29: Fraction of non-coding test sequences called non-coding, by dominant repeat class, full length, CDS vs intergenic.

model	L1 (845)	LTR (533)	none (304)	Alu (261)	other (228)	L2 (148)	DNA (137)	MIR (110)	simple (22)	low cx (2)
GROVER	0.994	0.987	0.980	1.000	0.991	0.980	0.964	1.000	0.955	1.000
DNABERT-2	1.000	0.992	0.974	1.000	1.000	0.993	1.000	0.991	1.000	1.000
NT-v2 50M	0.995	0.994	0.993	1.000	0.991	0.993	0.993	0.991	0.955	1.000
NT-v2 100M	0.994	0.994	0.997	1.000	0.991	1.000	1.000	1.000	1.000	1.000
NT-v2 250M	0.999	0.996	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000
NT-v2 500M	0.998	0.996	0.997	1.000	1.000	1.000	0.993	1.000	1.000	1.000
Hyena tiny-1k d128	0.988	0.964	0.970	1.000	0.978	0.986	0.956	0.909	1.000	1.000
Hyena tiny-1k d256	0.987	0.966	0.957	1.000	0.987	0.986	0.971	0.945	0.955	1.000
Hyena tiny-16k	0.991	0.949	0.951	1.000	0.982	0.986	0.978	0.945	1.000	1.000
Hyena small-32k	0.995	0.981	0.954	1.000	0.991	0.993	0.978	0.991	1.000	1.000
Hyena medium-160k	0.994	0.977	0.967	1.000	0.974	0.973	0.964	0.945	0.955	1.000
Hyena medium-450k	0.995	0.977	0.980	1.000	0.982	0.993	0.993	0.955	1.000	1.000
Hyena large-1m	0.993	0.970	0.954	1.000	0.974	0.986	0.985	0.964	1.000	1.000
Caduceus-Ph 1k d118	0.985	0.961	0.961	1.000	0.974	0.966	0.964	0.955	0.955	1.000
Caduceus-Ps 1k d118	0.992	0.962	0.964	1.000	0.978	0.986	0.956	1.000	1.000	1.000
Caduceus-Ph 1k d256	0.992	0.977	0.980	1.000	0.978	0.993	0.978	1.000	1.000	1.000
Caduceus-Ps 1k d256	0.995	0.989	0.980	1.000	0.987	1.000	0.978	1.000	0.955	1.000
Caduceus-Ph 131k	0.999	0.994	0.980	1.000	0.996	1.000	0.985	1.000	1.000	1.000
Caduceus-Ps 131k	0.999	0.998	0.987	1.000	0.996	1.000	0.971	1.000	1.000	1.000
Evo 1	0.992	0.983	0.990	0.996	0.982	1.000	0.985	1.000	0.955	1.000
Evo 2	0.998	0.985	0.990	1.000	0.961	1.000	0.985	0.991	0.955	1.000

Table 30: Spearman correlation of each feature with the coding score of non-coding test sequences, and 5-fold CV R^2 of a linear model from all features to the coding score, full length, CDS vs intergenic.

model	Alu	L1	MIR	L2	LTR	DNA	simple	low cx	other	GC	CpG o/e	homopol.	tandem	entropy	CV R^2
GROVER	-0.18	-0.51	0.13	0.15	0.12	0.15	0.01	0.06	-0.01	0.10	0.01	-0.14	-0.24	0.24	0.50
DNABERT-2	-0.13	-0.40	0.19	0.16	0.08	0.11	0.08	0.09	-0.18	0.09	-0.06	-0.00	-0.25	0.26	0.56
NT-v2 50M	-0.15	-0.44	-0.01	-0.03	0.05	-0.04	-0.13	-0.06	0.09	0.36	0.17	-0.37	-0.13	0.16	0.38
NT-v2 100M	-0.24	-0.41	-0.13	-0.09	-0.05	-0.10	-0.24	-0.05	0.22	0.25	0.19	-0.47	-0.05	-0.01	0.37
NT-v2 250M	-0.26	-0.46	-0.08	-0.06	-0.10	-0.08	-0.19	-0.03	0.20	0.19	0.15	-0.46	-0.07	-0.03	0.37
NT-v2 500M	-0.28	-0.40	-0.17	-0.15	-0.14	-0.11	-0.22	-0.05	0.18	0.13	0.17	-0.48	0.00	-0.13	0.35
Hyena tiny-1k d128	-0.43	-0.43	0.12	0.09	0.10	0.06	-0.13	0.00	0.21	0.18	0.02	-0.31	-0.30	0.18	0.44
Hyena tiny-1k d256	-0.38	-0.43	0.10	0.09	0.05	0.04	-0.14	0.01	0.22	0.13	0.04	-0.32	-0.26	0.15	0.36
Hyena tiny-16k	-0.48	-0.42	0.08	0.07	0.07	0.03	-0.15	-0.01	0.24	0.16	0.03	-0.38	-0.25	0.11	0.45
Hyena small-32k	-0.29	-0.56	0.09	0.12	-0.03	0.04	-0.02	0.05	0.26	0.04	0.10	-0.24	-0.12	0.12	0.54
Hyena medium-160k	-0.27	-0.54	0.11	0.14	0.02	0.03	-0.10	0.02	0.27	0.18	0.11	-0.29	-0.24	0.21	0.50
Hyena medium-450k	-0.33	-0.49	0.13	0.13	0.03	0.04	-0.06	0.04	0.27	0.09	0.10	-0.29	-0.22	0.15	0.46
Hyena large-1m	-0.36	-0.36	0.12	0.11	0.08	0.04	-0.08	0.02	0.01	0.07	-0.06	-0.29	-0.22	0.09	0.40
Caduceus-Ph 1k d118	-0.31	-0.54	0.09	0.09	0.05	0.06	-0.12	0.00	0.29	0.24	0.13	-0.29	-0.25	0.21	0.49
Caduceus-Ps 1k d118	-0.27	-0.56	-0.03	0.08	0.09	0.06	-0.12	0.01	0.28	0.24	0.13	-0.29	-0.22	0.22	0.50
Caduceus-Ph 1k d256	-0.22	-0.56	-0.01	0.09	0.06	0.08	-0.08	0.01	0.23	0.20	0.11	-0.25	-0.23	0.21	0.49
Caduceus-Ps 1k d256	-0.25	-0.53	-0.01	0.09	0.14	0.10	-0.06	0.03	0.15	0.15	0.05	-0.20	-0.25	0.23	0.53
Caduceus-Ph 131k	-0.12	-0.58	0.04	0.04	0.08	0.07	0.02	0.07	0.07	0.09	0.01	-0.13	-0.18	0.22	0.57
Caduceus-Ps 131k	-0.06	-0.44	0.02	-0.07	0.14	0.10	0.04	0.05	-0.04	0.15	-0.05	-0.06	-0.20	0.26	0.47
Evo 1	-0.20	-0.25	-0.02	-0.05	0.06	-0.11	-0.30	-0.10	0.19	0.51	0.13	-0.44	-0.31	0.17	0.48
Evo 2	-0.03	-0.41	0.18	0.11	0.13	0.06	0.01	0.09	-0.03	0.36	0.10	-0.07	-0.32	0.39	0.45

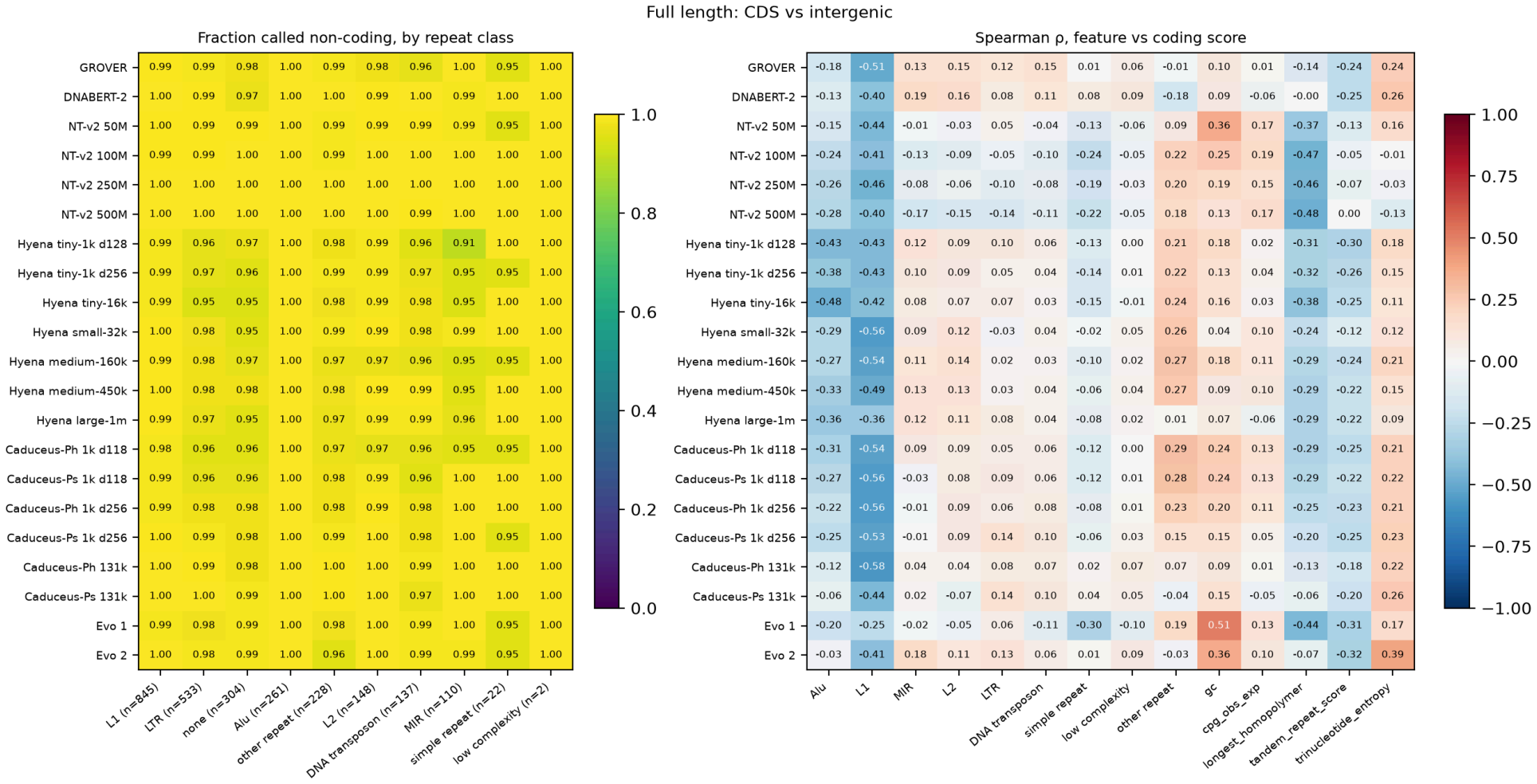


Figure 15: Fraction called non-coding by repeat class (left) and feature correlations with the coding score (right), full length, CDS vs intergenic.

Table 31: Fraction of non-coding test sequences called non-coding, by dominant repeat class, full length, CDS vs lncRNA.

model	none (684)	LTR (617)	L1 (306)	Alu (259)	DNA (149)	MIR (130)	L2 (101)	other (49)	simple (11)	low cx (3)
GROVER	0.909	0.969	0.984	0.996	0.953	0.992	0.960	0.898	1.000	0.667
DNABERT-2	0.937	0.985	0.990	1.000	0.973	0.977	0.950	0.959	1.000	1.000
NT-v2 50M	0.984	0.997	0.997	0.988	0.993	1.000	1.000	0.980	1.000	1.000
NT-v2 100M	0.982	0.994	1.000	0.996	0.993	1.000	1.000	0.980	1.000	1.000
NT-v2 250M	0.985	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
NT-v2 500M	0.985	0.995	0.997	0.996	1.000	1.000	1.000	0.959	1.000	1.000
Hyena tiny-1k d128	0.879	0.916	0.964	0.996	0.926	0.946	0.911	0.878	1.000	0.667
Hyena tiny-1k d256	0.898	0.925	0.961	0.996	0.933	0.954	0.921	0.898	1.000	0.333
Hyena tiny-16k	0.864	0.917	0.938	1.000	0.886	0.923	0.931	0.837	0.909	0.667
Hyena small-32k	0.901	0.969	0.971	1.000	0.973	0.992	0.960	0.898	1.000	0.667
Hyena medium-160k	0.899	0.956	0.967	1.000	0.960	0.969	0.941	0.898	1.000	0.667
Hyena medium-450k	0.892	0.958	0.967	0.992	0.953	0.938	0.960	0.898	0.909	0.667
Hyena large-1m	0.898	0.948	0.971	0.992	0.926	0.962	0.960	0.898	0.909	1.000
Caduceus-Ph 1k d118	0.882	0.947	0.954	0.996	0.913	0.962	0.941	0.816	1.000	0.667
Caduceus-Ps 1k d118	0.879	0.942	0.984	1.000	0.940	0.977	0.950	0.898	1.000	0.333
Caduceus-Ph 1k d256	0.899	0.966	0.984	1.000	0.973	0.992	0.950	0.898	1.000	0.667
Caduceus-Ps 1k d256	0.908	0.977	0.987	0.996	0.987	1.000	0.990	0.918	1.000	1.000
Caduceus-Ph 131k	0.917	0.982	1.000	1.000	0.987	0.992	0.990	0.939	1.000	1.000
Caduceus-Ps 131k	0.947	0.994	0.997	1.000	0.993	0.992	0.980	1.000	1.000	1.000
Evo 1	0.975	0.995	0.990	0.992	1.000	0.992	1.000	1.000	1.000	1.000
Evo 2	0.980	0.998	0.964	0.996	1.000	1.000	1.000	1.000	1.000	1.000

Table 32: Spearman correlation of each feature with the coding score of non-coding test sequences, and 5-fold CV R^2 of a linear model from all features to the coding score, full length, CDS vs lncRNA.

model	Alu	L1	MIR	L2	LTR	DNA	simple	low cx	other	GC	CpG o/e	homopol.	tandem	entropy	CV R^2
GROVER	-0.44	-0.30	-0.10	-0.06	-0.19	-0.01	-0.09	0.00	0.01	0.16	0.07	-0.27	-0.15	-0.02	0.46
DNABERT-2	-0.35	-0.34	-0.09	-0.10	-0.30	-0.08	-0.04	0.03	-0.01	0.20	0.09	-0.22	-0.11	-0.01	0.42
NT-v2 50M	-0.27	-0.20	-0.13	-0.10	-0.04	-0.09	-0.12	-0.04	-0.04	0.28	0.12	-0.36	0.03	-0.21	-0.12
NT-v2 100M	-0.31	-0.22	-0.18	-0.17	-0.10	-0.15	-0.21	-0.08	-0.06	0.25	0.09	-0.46	-0.03	-0.24	-0.63
NT-v2 250M	-0.30	-0.18	-0.19	-0.16	-0.08	-0.14	-0.18	-0.06	-0.06	0.20	0.04	-0.42	0.01	-0.29	-0.89
NT-v2 500M	-0.26	-0.16	-0.21	-0.14	-0.09	-0.13	-0.22	-0.08	-0.03	0.14	0.02	-0.42	-0.04	-0.27	-0.35
Hyena tiny-1k d128	-0.61	-0.23	-0.00	-0.01	-0.02	-0.03	-0.13	-0.04	0.01	0.23	0.06	-0.41	-0.28	0.01	0.66
Hyena tiny-1k d256	-0.58	-0.20	-0.00	0.02	-0.08	-0.01	-0.13	-0.02	0.01	0.20	0.00	-0.39	-0.28	-0.01	0.62
Hyena tiny-16k	-0.64	-0.23	-0.01	-0.01	-0.07	-0.02	-0.14	-0.03	-0.00	0.22	0.02	-0.43	-0.25	-0.04	0.59
Hyena small-32k	-0.50	-0.28	-0.08	-0.02	-0.27	-0.05	-0.07	0.03	0.02	0.14	0.04	-0.30	-0.12	-0.07	0.49
Hyena medium-160k	-0.56	-0.29	-0.04	-0.02	-0.17	-0.04	-0.14	0.00	0.01	0.21	0.02	-0.39	-0.21	-0.08	0.54
Hyena medium-450k	-0.53	-0.25	-0.03	-0.03	-0.14	-0.06	-0.13	-0.01	0.01	0.21	0.03	-0.41	-0.24	-0.08	0.44
Hyena large-1m	-0.60	-0.26	-0.07	-0.06	-0.10	-0.05	-0.17	-0.01	-0.00	0.17	0.01	-0.44	-0.17	-0.15	0.52
Caduceus-Ph 1k d118	-0.52	-0.21	-0.10	-0.06	-0.12	-0.01	-0.15	-0.02	0.00	0.18	0.08	-0.31	-0.20	-0.01	0.39
Caduceus-Ps 1k d118	-0.51	-0.25	-0.15	-0.09	-0.14	-0.03	-0.15	-0.01	0.01	0.20	0.04	-0.38	-0.19	-0.02	0.53
Caduceus-Ph 1k d256	-0.48	-0.28	-0.18	-0.08	-0.14	-0.01	-0.11	-0.01	-0.00	0.18	0.03	-0.33	-0.19	-0.02	0.49
Caduceus-Ps 1k d256	-0.45	-0.29	-0.21	-0.12	-0.20	-0.03	-0.11	0.01	0.01	0.14	0.06	-0.30	-0.15	-0.02	0.50
Caduceus-Ph 131k	-0.41	-0.37	-0.22	-0.21	-0.29	-0.15	-0.06	0.00	-0.02	0.14	0.06	-0.31	-0.02	-0.15	0.44
Caduceus-Ps 131k	-0.36	-0.36	-0.23	-0.19	-0.30	-0.14	-0.06	0.01	-0.01	0.11	0.00	-0.26	-0.05	-0.14	0.51
Evo 1	-0.17	-0.17	-0.13	-0.11	-0.14	-0.15	-0.14	-0.04	-0.05	0.42	0.18	-0.40	0.05	-0.21	0.19
Evo 2	0.08	0.12	0.04	0.07	0.14	0.07	-0.04	-0.01	-0.04	0.27	-0.06	0.04	-0.11	-0.04	0.28

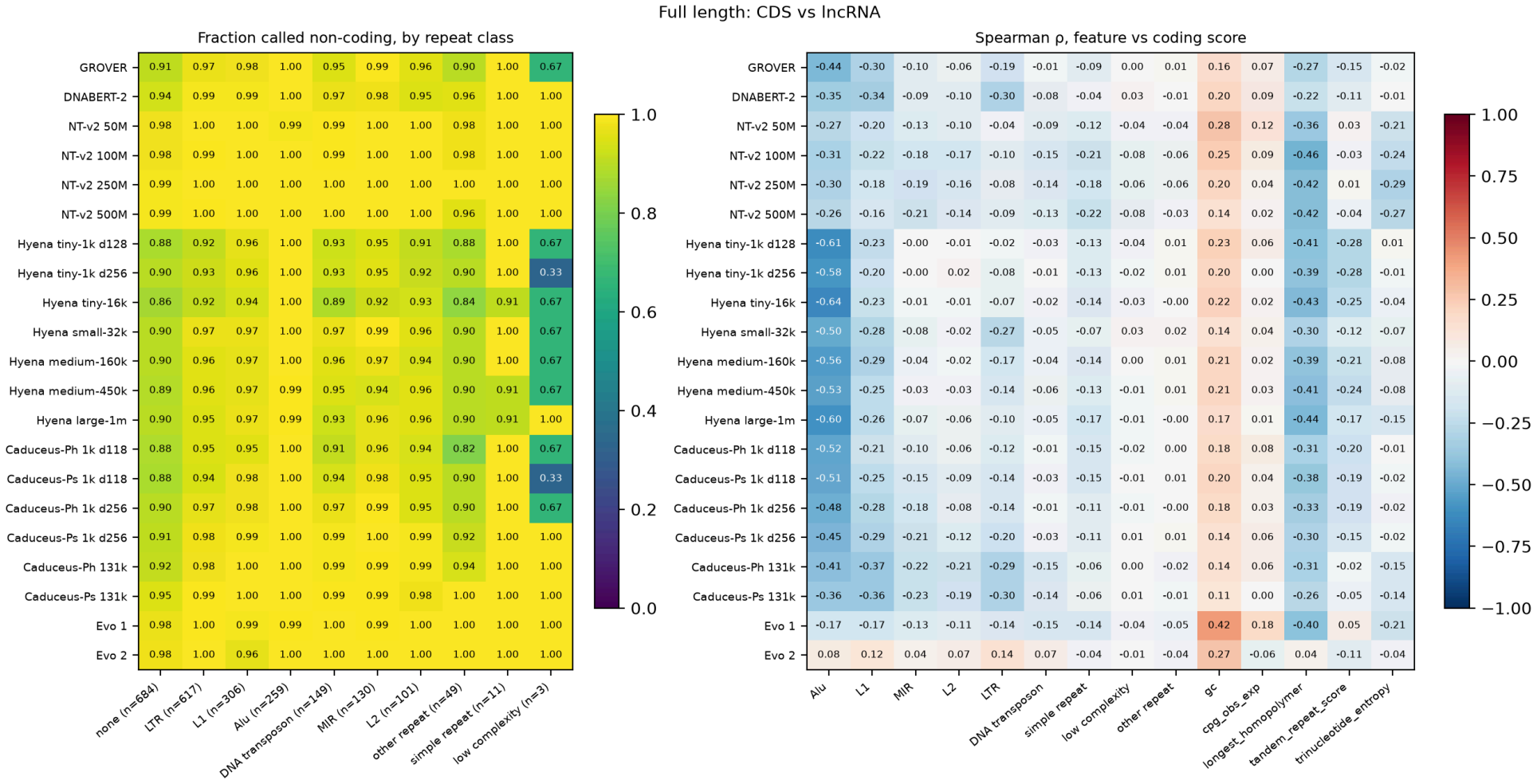


Figure 16: Fraction called non-coding by repeat class (left) and feature correlations with the coding score (right), full length, CDS vs lncRNA.

Table 33: Fraction of non-coding test sequences called non-coding, by dominant repeat class, full length, CDS vs intron.

model	none (621)	Alu (571)	MIR (169)	L1 (123)	DNA (118)	L2 (84)	LTR (68)	other (26)	simple (23)	low cx (5)
GROVER	0.973	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.957	1.000
DNABERT-2	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
NT-v2 50M	0.992	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
NT-v2 100M	0.992	1.000	1.000	1.000	0.992	1.000	1.000	1.000	1.000	1.000
NT-v2 250M	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
NT-v2 500M	0.994	1.000	1.000	1.000	0.992	1.000	1.000	1.000	1.000	1.000
Hyena tiny-1k d128	0.965	0.998	0.970	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Hyena tiny-1k d256	0.976	1.000	0.982	1.000	0.975	0.988	1.000	1.000	1.000	1.000
Hyena tiny-16k	0.950	1.000	0.964	0.984	0.966	0.988	0.971	0.962	1.000	1.000
Hyena small-32k	0.977	1.000	1.000	1.000	1.000	1.000	1.000	0.962	1.000	1.000
Hyena medium-160k	0.976	1.000	0.976	1.000	0.992	1.000	1.000	0.962	1.000	1.000
Hyena medium-450k	0.981	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Hyena large-1m	0.969	1.000	0.994	1.000	0.992	1.000	1.000	1.000	0.957	1.000
Caduceus-Ph 1k d118	0.966	1.000	0.988	0.992	0.992	0.988	1.000	1.000	1.000	1.000
Caduceus-Ps 1k d118	0.968	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Caduceus-Ph 1k d256	0.974	1.000	1.000	1.000	1.000	1.000	0.985	1.000	1.000	1.000
Caduceus-Ps 1k d256	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Caduceus-Ph 131k	0.994	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Caduceus-Ps 131k	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Evo 1	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Evo 2	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Table 34: Spearman correlation of each feature with the coding score of non-coding test sequences, and 5-fold CV R^2 of a linear model from all features to the coding score, full length, CDS vs intron.

model	Alu	L1	MIR	L2	LTR	DNA	simple	low cx	other	GC	CpG o/e	homopol.	tandem	entropy	CV R^2
GROVER	-0.43	-0.19	-0.12	-0.07	0.02	0.04	-0.06	-0.05	0.07	0.22	-0.02	-0.30	-0.24	0.04	0.43
DNABERT-2	-0.46	-0.19	-0.05	-0.06	0.01	-0.05	-0.07	-0.06	0.03	0.20	-0.04	-0.33	-0.23	0.02	0.38
NT-v2 50M	-0.39	-0.31	-0.11	-0.18	-0.17	-0.21	-0.13	-0.09	-0.12	0.44	0.11	-0.48	0.00	-0.15	-0.25
NT-v2 100M	-0.44	-0.33	-0.23	-0.27	-0.20	-0.21	-0.19	-0.09	-0.11	0.16	0.05	-0.48	0.10	-0.25	-0.60
NT-v2 250M	-0.55	-0.34	-0.22	-0.23	-0.21	-0.23	-0.19	-0.11	-0.11	0.24	0.00	-0.57	0.06	-0.31	-0.86
NT-v2 500M	-0.42	-0.29	-0.29	-0.25	-0.20	-0.24	-0.22	-0.13	-0.09	0.18	0.07	-0.48	0.07	-0.25	-0.60
Hyena tiny-1k d128	-0.71	-0.19	0.08	0.02	0.00	-0.03	-0.14	-0.06	-0.01	0.25	-0.09	-0.53	-0.26	-0.03	0.73
Hyena tiny-1k d256	-0.62	-0.15	0.08	0.04	0.01	-0.00	-0.12	-0.07	0.02	0.26	-0.05	-0.46	-0.32	0.04	0.66
Hyena tiny-16k	-0.77	-0.22	0.05	-0.02	-0.03	-0.06	-0.15	-0.07	-0.03	0.25	-0.14	-0.59	-0.22	-0.12	0.75
Hyena small-32k	-0.60	-0.27	-0.18	-0.11	-0.10	-0.11	-0.10	-0.05	-0.01	0.20	-0.03	-0.48	-0.11	-0.12	0.51
Hyena medium-160k	-0.69	-0.27	-0.01	-0.02	-0.11	-0.09	-0.19	-0.09	-0.01	0.21	-0.08	-0.55	-0.19	-0.12	0.57
Hyena medium-450k	-0.61	-0.19	-0.00	0.01	-0.02	-0.05	-0.14	-0.08	0.02	0.16	-0.09	-0.46	-0.25	-0.05	0.55
Hyena large-1m	-0.79	-0.21	0.02	-0.02	-0.07	-0.07	-0.16	-0.09	-0.01	0.18	-0.14	-0.60	-0.16	-0.17	0.76
Caduceus-Ph 1k d118	-0.60	-0.17	-0.05	-0.03	-0.04	-0.00	-0.15	-0.09	0.02	0.13	-0.06	-0.40	-0.24	-0.03	0.61
Caduceus-Ps 1k d118	-0.56	-0.17	-0.27	-0.08	-0.05	-0.03	-0.18	-0.10	0.01	0.16	-0.04	-0.44	-0.16	-0.08	0.58
Caduceus-Ph 1k d256	-0.48	-0.31	-0.18	-0.08	-0.05	0.01	-0.15	-0.08	0.04	0.16	-0.06	-0.40	-0.24	-0.02	0.56
Caduceus-Ps 1k d256	-0.56	-0.22	-0.30	-0.09	-0.01	0.00	-0.15	-0.08	0.03	0.10	-0.10	-0.41	-0.18	-0.10	0.64
Caduceus-Ph 131k	-0.54	-0.37	-0.27	-0.21	-0.09	-0.22	-0.10	-0.07	-0.02	0.22	0.03	-0.44	-0.03	-0.15	0.55
Caduceus-Ps 131k	-0.46	-0.25	-0.34	-0.27	-0.01	-0.11	-0.07	-0.07	0.01	0.10	-0.06	-0.36	0.00	-0.19	0.43
Evo 1	-0.27	-0.19	-0.05	-0.09	-0.13	-0.15	-0.13	-0.07	-0.06	0.41	0.09	-0.45	-0.05	-0.08	0.13
Evo 2	0.12	-0.05	0.07	-0.01	0.06	0.05	0.00	0.00	0.02	0.22	-0.06	0.01	-0.21	0.15	0.09

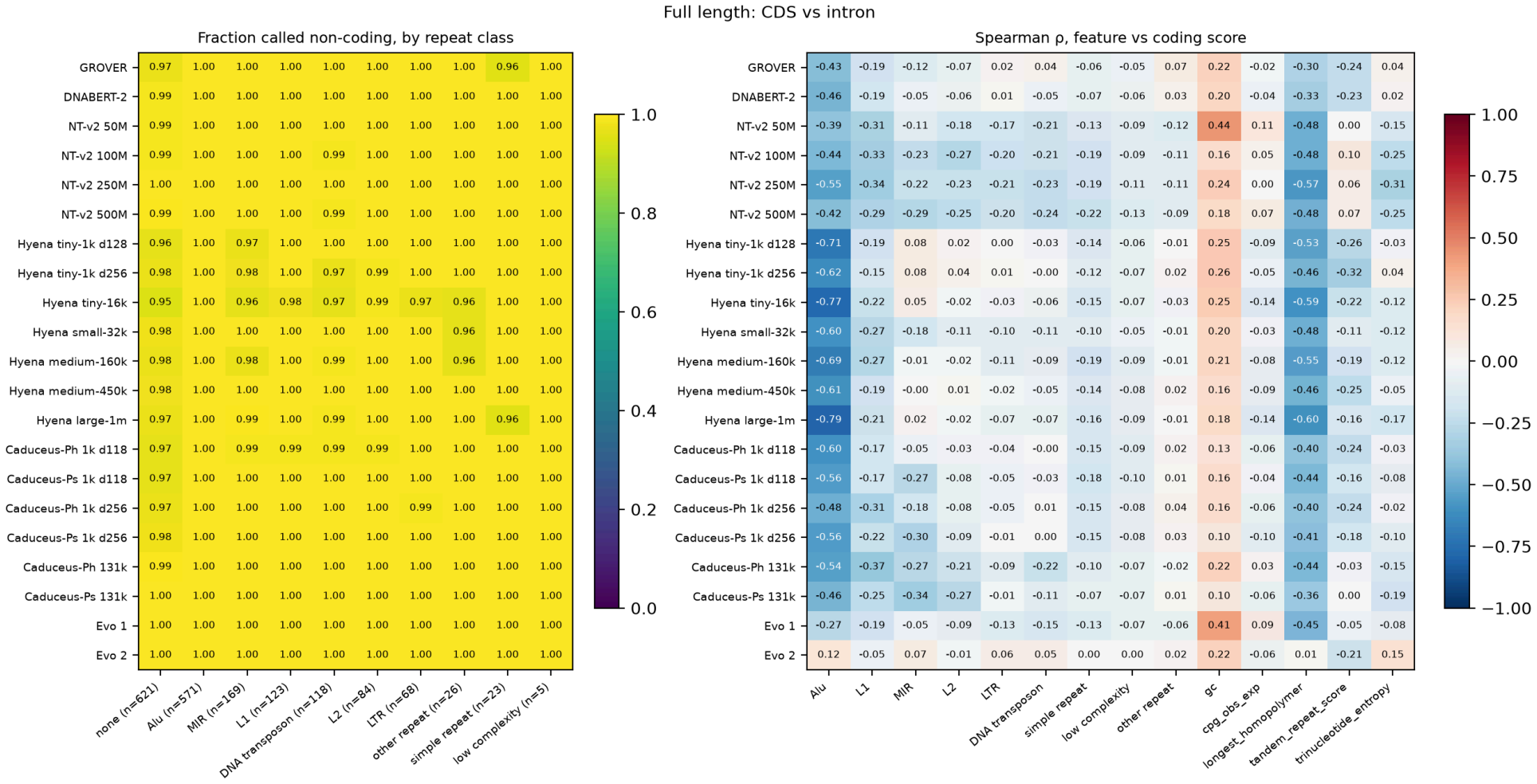


Figure 17: Fraction called non-coding by repeat class (left) and feature correlations with the coding score (right), full length, CDS vs intron.

8 Robustness to training-set size

8.1 Rationale

The full-length sets were capped at 10,000 sequences per class, and every experiment uses a probe trained on all available training sequences. This section measures how probe accuracy and the probe itself change when the probe is trained on fewer sequences. If both are already stable well below the sizes used, the results do not depend on the choice of cap.

8.2 Method

- **Reference probe:** the standard probe trained on the whole training set, as in Experiments A–D (200 bp: 75,000 windows; full length: 15,575–16,219 sequences, 7,410–8,165 per class).
- **Subset probes:** the same probe trained on balanced random subsets of n training sequences ($n/2$ per class), for n = 500, 1,000, 2,000, 5,000 and 10,000, plus 20,000 and 50,000 for 200 bp, plus the largest balanced subset available (twice the size of the smaller class). Ten subsets are drawn per size (seeds 0–9); a subset equal to the whole training set is fitted once.
- Each subset probe is evaluated on the unchanged full test set and compared with the reference probe:
  - **test accuracy** (balanced accuracy for full length);
  - **Pearson correlation of coding scores** with the reference scores over the test set. For two linear probes this is the cosine similarity of their directions measured under the covariance of the test embeddings, so it compares the probes along directions in which the embeddings actually vary;
  - **Spearman correlation** of the scores, and the **fraction of identical calls** (coding / non-coding);
  - **cosine similarity of the weight vectors** in raw embedding space (logistic-regression coefficients divided by the standardisation scale). Embedding dimensions are strongly correlated, so different weight vectors can produce nearly identical scores; this metric is reported for completeness.
- Script: `sample_size.py` (cached embeddings only). Full per-run results: `results/sample_size.csv`; mean and SD over seeds: `results/sample_size_summary.csv`.

8.3 Results

Table columns: *all data* = accuracy of the reference probe; *1k*, *10k* = mean test accuracy of subset probes trained on 1,000 or 10,000 sequences; *10k SD* = its standard deviation over the 10 subsets; *Pearson*, *same call* and *cosine* = agreement with the reference probe, averaged over subsets.

Table 35: Probe accuracy and agreement with the reference probe by training-set size, 200 bp Genomic Benchmarks.

model	all data	1k	10k	10k SD	Pearson 1k	Pearson 10k	same call 10k	cosine 10k
GROVER	0.906	0.860	0.888	0.002	0.898	0.961	0.945	0.547
DNABERT-2	0.912	0.858	0.893	0.001	0.897	0.966	0.944	0.697
NT-v2 50M	0.893	0.839	0.882	0.001	0.898	0.981	0.953	0.760
NT-v2 100M	0.886	0.827	0.876	0.001	0.885	0.981	0.951	0.788
NT-v2 250M	0.907	0.854	0.891	0.002	0.906	0.974	0.947	0.704
NT-v2 500M	0.892	0.832	0.869	0.002	0.886	0.962	0.927	0.660
Hyena tiny-1k d128	0.866	0.846	0.864	0.001	0.949	0.993	0.973	0.934
Hyena tiny-1k d256	0.879	0.841	0.872	0.001	0.920	0.988	0.962	0.869
Hyena tiny-16k	0.864	0.842	0.859	0.001	0.947	0.992	0.973	0.924
Hyena small-32k	0.879	0.845	0.873	0.001	0.924	0.988	0.964	0.864
Hyena medium-160k	0.881	0.857	0.877	0.001	0.937	0.989	0.968	0.863
Hyena medium-450k	0.868	0.832	0.863	0.001	0.911	0.986	0.961	0.829
Hyena large-1m	0.873	0.842	0.867	0.001	0.921	0.989	0.967	0.874
Caduceus-Ph 1k d118	0.874	0.857	0.872	0.001	0.932	0.989	0.977	0.881
Caduceus-Ps 1k d118	0.887	0.860	0.881	0.001	0.883	0.970	0.972	0.829
Caduceus-Ph 1k d256	0.900	0.876	0.895	0.001	0.929	0.981	0.972	0.890
Caduceus-Ps 1k d256	0.911	0.883	0.904	0.001	0.927	0.974	0.968	0.641
Caduceus-Ph 131k	0.923	0.906	0.919	0.001	0.954	0.987	0.979	0.076
Caduceus-Ps 131k	0.936	0.914	0.931	0.001	0.934	0.977	0.976	0.627
Evo 1	0.893	0.878	0.888	0.001	0.938	0.990	0.974	0.775
Evo 2	0.932	0.917	0.926	0.001	0.979	0.996	0.985	0.929

Table 36: Probe accuracy and agreement with the reference probe by training-set size, full length, CDS vs intergenic.

model	all data	1k	10k	10k SD	Pearson 1k	Pearson 10k	same call 10k	cosine 10k
GROVER	0.988	0.985	0.989	0.001	0.964	0.988	0.994	0.808
DNABERT-2	0.994	0.990	0.994	0.001	0.977	0.993	0.996	0.882
NT-v2 50M	0.996	0.993	0.996	0.001	0.972	0.996	0.998	0.915
NT-v2 100M	0.996	0.990	0.995	0.000	0.978	0.996	0.998	0.924
NT-v2 250M	0.998	0.993	0.997	0.001	0.972	0.996	0.999	0.939
NT-v2 500M	0.996	0.989	0.995	0.000	0.961	0.996	0.997	0.911
Hyena tiny-1k d128	0.980	0.962	0.979	0.001	0.977	0.997	0.997	0.980
Hyena tiny-1k d256	0.979	0.968	0.979	0.001	0.971	0.997	0.994	0.945
Hyena tiny-16k	0.977	0.960	0.976	0.001	0.964	0.996	0.995	0.974
Hyena small-32k	0.986	0.972	0.985	0.001	0.974	0.996	0.995	0.950
Hyena medium-160k	0.979	0.968	0.978	0.001	0.962	0.995	0.996	0.945
Hyena medium-450k	0.987	0.973	0.985	0.001	0.969	0.996	0.995	0.939
Hyena large-1m	0.982	0.967	0.980	0.001	0.961	0.995	0.995	0.954
Caduceus-Ph 1k d118	0.974	0.952	0.969	0.001	0.937	0.995	0.994	0.969
Caduceus-Ps 1k d118	0.981	0.962	0.977	0.001	0.957	0.996	0.995	0.967
Caduceus-Ph 1k d256	0.989	0.978	0.987	0.001	0.954	0.996	0.997	0.958
Caduceus-Ps 1k d256	0.993	0.985	0.993	0.001	0.961	0.991	0.997	0.917
Caduceus-Ph 131k	0.993	0.989	0.992	0.001	0.984	0.996	0.998	0.873
Caduceus-Ps 131k	0.996	0.993	0.996	0.000	0.965	0.991	0.998	0.807
Evo 1	0.991	0.987	0.991	0.000	0.975	0.998	0.998	0.957
Evo 2	0.991	0.988	0.990	0.001	0.983	0.999	0.999	0.999

Table 37: Probe accuracy and agreement with the reference probe by training-set size, full length, CDS vs lncRNA.

model	all data	1k	10k	10k SD	Pearson 1k	Pearson 10k	same call 10k	cosine 10k
GROVER	0.953	0.924	0.949	0.001	0.921	0.988	0.977	0.859
DNABERT-2	0.968	0.957	0.962	0.002	0.911	0.979	0.977	0.811
NT-v2 50M	0.993	0.981	0.991	0.001	0.944	0.991	0.994	0.878
NT-v2 100M	0.991	0.980	0.989	0.001	0.933	0.993	0.995	0.905
NT-v2 250M	0.993	0.982	0.992	0.002	0.952	0.990	0.995	0.899
NT-v2 500M	0.991	0.977	0.989	0.001	0.934	0.990	0.995	0.881
Hyena tiny-1k d128	0.927	0.906	0.927	0.002	0.979	0.998	0.986	0.986
Hyena tiny-1k d256	0.937	0.907	0.935	0.002	0.961	0.997	0.985	0.966
Hyena tiny-16k	0.922	0.904	0.922	0.002	0.984	0.998	0.988	0.978
Hyena small-32k	0.956	0.921	0.952	0.001	0.966	0.997	0.987	0.960
Hyena medium-160k	0.949	0.925	0.948	0.002	0.968	0.997	0.988	0.968
Hyena medium-450k	0.942	0.913	0.941	0.001	0.965	0.998	0.987	0.969
Hyena large-1m	0.946	0.917	0.945	0.002	0.970	0.998	0.988	0.966
Caduceus-Ph 1k d118	0.936	0.910	0.934	0.002	0.965	0.998	0.990	0.984
Caduceus-Ps 1k d118	0.947	0.921	0.946	0.002	0.944	0.996	0.992	0.957
Caduceus-Ph 1k d256	0.959	0.938	0.956	0.002	0.962	0.995	0.993	0.978
Caduceus-Ps 1k d256	0.968	0.947	0.967	0.001	0.953	0.993	0.990	0.927
Caduceus-Ph 131k	0.971	0.959	0.970	0.001	0.962	0.995	0.994	0.902
Caduceus-Ps 131k	0.983	0.972	0.982	0.001	0.943	0.991	0.995	0.939
Evo 1	0.987	0.983	0.986	0.000	0.979	0.999	0.999	0.897
Evo 2	0.984	0.980	0.983	0.000	0.984	0.999	0.998	0.997

Table 38: Probe accuracy and agreement with the reference probe by training-set size, full length, CDS vs intron.

model	all data	1k	10k	10k SD	Pearson 1k	Pearson 10k	same call 10k	cosine 10k
GROVER	0.990	0.988	0.991	0.001	0.972	0.993	0.995	0.824
DNABERT-2	0.996	0.990	0.996	0.001	0.975	0.996	0.997	0.921
NT-v2 50M	0.998	0.994	0.998	0.001	0.970	0.996	0.998	0.892
NT-v2 100M	0.997	0.994	0.997	0.000	0.976	0.995	0.998	0.907
NT-v2 250M	0.999	0.995	0.999	0.000	0.975	0.997	0.999	0.940
NT-v2 500M	0.997	0.993	0.997	0.001	0.967	0.996	0.998	0.906
Hyena tiny-1k d128	0.987	0.975	0.984	0.001	0.977	0.997	0.996	0.973
Hyena tiny-1k d256	0.988	0.980	0.987	0.001	0.974	0.996	0.996	0.935
Hyena tiny-16k	0.979	0.971	0.979	0.001	0.981	0.998	0.995	0.973
Hyena small-32k	0.989	0.977	0.988	0.001	0.959	0.994	0.994	0.937
Hyena medium-160k	0.987	0.974	0.986	0.001	0.970	0.997	0.996	0.954
Hyena medium-450k	0.992	0.981	0.991	0.001	0.976	0.996	0.996	0.933
Hyena large-1m	0.989	0.972	0.988	0.001	0.976	0.996	0.996	0.956
Caduceus-Ph 1k d118	0.986	0.972	0.985	0.001	0.963	0.996	0.998	0.999
Caduceus-Ps 1k d118	0.990	0.976	0.988	0.001	0.960	0.997	0.997	0.939
Caduceus-Ph 1k d256	0.991	0.982	0.990	0.001	0.949	0.995	0.998	0.935
Caduceus-Ps 1k d256	0.994	0.986	0.993	0.001	0.968	0.995	0.997	0.871
Caduceus-Ph 131k	0.995	0.990	0.994	0.001	0.977	0.997	0.998	0.988
Caduceus-Ps 131k	0.997	0.993	0.996	0.001	0.977	0.994	0.998	0.846
Evo 1	0.996	0.991	0.995	0.000	0.978	0.998	0.999	0.945
Evo 2	0.995	0.992	0.994	0.000	0.974	0.998	1.000	0.998

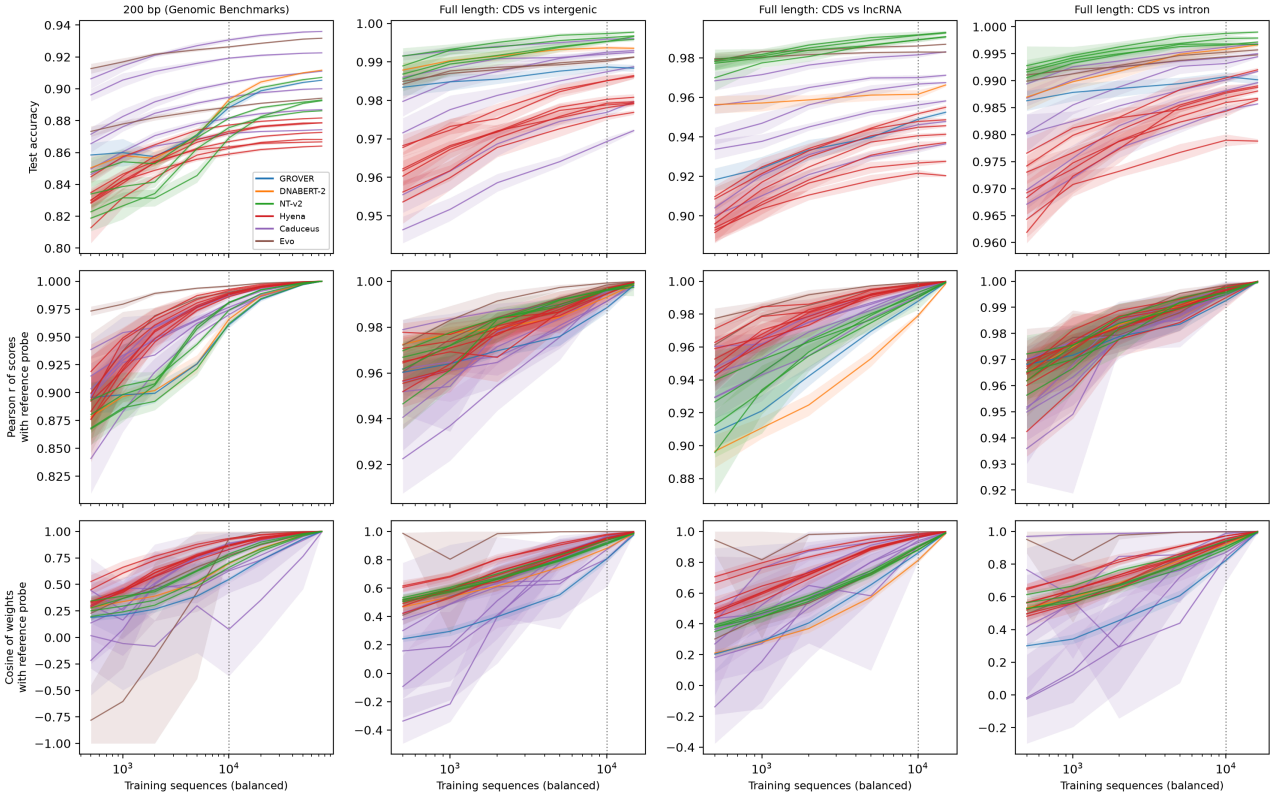


Figure 18: Test accuracy (top), Pearson correlation of coding scores with the reference probe (middle) and cosine similarity of probe weights with the reference probe (bottom) against the number of training sequences. One line per model, coloured by family; bands show ±1 SD over 10 subsets. Dotted line: 10,000 sequences.

Appendix

A.1 Mean coding scores under each edit

Table 39: Mean coding score of edited **coding** sequences, 200 bp Genomic Benchmarks.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	3.336	3.292	1.925	3.898	4.102	2.609	2.730	1.815
DNABERT-2	3.828	3.073	1.708	4.318	4.307	2.645	3.123	2.070
NT-v2 50M	3.449	1.793	0.736	3.542	3.410	2.046	2.900	2.066
NT-v2 100M	3.103	2.190	0.844	3.397	3.446	1.768	2.547	1.774
NT-v2 250M	4.053	1.692	0.522	4.301	4.015	1.641	3.433	2.287
NT-v2 500M	3.407	2.580	1.168	3.798	3.802	2.109	2.771	2.139
Hyena tiny-1k d128	2.269	3.144	1.733	2.958	3.172	1.886	1.931	1.192
Hyena tiny-1k d256	2.569	3.384	1.762	3.318	3.620	2.135	2.166	1.362
Hyena tiny-16k	2.114	2.834	1.739	2.658	2.694	1.808	1.851	1.178
Hyena small-32k	2.634	3.272	1.929	3.175	3.368	2.302	2.398	1.483
Hyena medium-160k	2.615	2.902	1.722	2.930	3.227	2.136	2.279	1.408
Hyena medium-450k	2.428	3.253	1.781	3.210	3.365	2.003	2.060	1.442
Hyena large-1m	2.498	2.869	1.531	3.064	3.266	1.952	2.033	1.242
Caduceus-Ph 1k d118	2.404	3.260	1.707	3.075	3.485	2.124	2.229	1.485
Caduceus-Ps 1k d118	2.611	3.644	1.931	3.367	3.745	2.254	2.378	1.593
Caduceus-Ph 1k d256	3.388	3.007	1.586	3.805	4.063	2.763	3.016	2.176
Caduceus-Ps 1k d256	3.744	3.564	1.832	4.353	4.653	2.984	3.263	2.313
Caduceus-Ph 131k	5.251	4.312	2.185	5.781	6.412	4.198	4.995	3.190
Caduceus-Ps 131k	6.199	3.579	1.695	6.306	6.501	3.936	5.758	1.870
Evo 1	3.211	-0.057	-0.152	2.435	2.416	0.271	2.804	1.639
Evo 2	12.153	0.463	0.041	0.859	5.971	2.371	6.731	10.547

Table 40: Mean coding score of edited **non-coding** sequences, 200 bp Genomic Benchmarks.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	-3.959	0.711	-1.460	-0.578	1.392	-4.103	-4.093	-3.880
DNABERT-2	-4.393	0.282	-1.705	-0.708	1.431	-4.349	-4.382	-4.124
NT-v2 50M	-3.025	-0.575	-2.430	-1.560	0.498	-3.202	-3.090	-2.967
NT-v2 100M	-2.901	-0.340	-2.375	-1.421	0.570	-3.125	-3.083	-2.822
NT-v2 250M	-3.555	-1.090	-2.955	-1.986	0.238	-3.734	-3.743	-3.488
NT-v2 500M	-3.461	-0.516	-2.490	-1.597	0.686	-3.582	-3.602	-3.346
Hyena tiny-1k d128	-3.081	-0.124	-1.608	-1.084	0.789	-3.064	-3.068	-2.985
Hyena tiny-1k d256	-3.324	0.085	-1.760	-1.064	1.033	-3.320	-3.317	-3.205
Hyena tiny-16k	-3.399	-0.566	-1.664	-1.289	0.535	-3.314	-3.271	-3.373
Hyena small-32k	-3.987	0.066	-1.329	-0.818	0.899	-3.732	-3.729	-3.952
Hyena medium-160k	-3.414	-0.478	-1.751	-1.254	0.464	-3.337	-3.339	-3.386
Hyena medium-450k	-2.691	-0.421	-1.851	-1.302	0.789	-2.736	-2.728	-2.656
Hyena large-1m	-2.928	-0.449	-2.130	-1.466	0.694	-3.045	-3.031	-2.874
Caduceus-Ph 1k d118	-3.799	0.062	-1.454	-0.839	0.731	-3.715	-3.685	-3.809
Caduceus-Ps 1k d118	-4.878	0.301	-1.432	-0.758	0.576	-4.804	-4.780	-4.844
Caduceus-Ph 1k d256	-4.597	0.271	-1.613	-0.792	0.556	-4.512	-4.511	-4.540
Caduceus-Ps 1k d256	-4.829	0.409	-1.575	-0.752	0.763	-4.926	-4.889	-4.751
Caduceus-Ph 131k	-4.663	1.326	-1.042	-0.169	1.225	-4.558	-4.516	-4.679
Caduceus-Ps 131k	-4.917	0.795	-1.255	-0.491	0.650	-4.823	-4.794	-4.812
Evo 1	-2.822	-2.533	-2.836	-2.662	-1.503	-2.976	-2.987	-2.832
Evo 2	-3.407	-1.436	-2.062	-1.769	-0.954	-3.279	-3.278	-3.379

Table 41: Mean coding score of edited **coding** sequences, full length, CDS vs intergenic.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	16.949	10.041	9.121	17.133	15.824	13.092	13.676	9.093
DNABERT-2	20.207	5.905	7.593	19.704	17.955	13.715	18.139	14.916
NT-v2 50M	20.823	-3.648	-4.689	18.744	16.921	5.801	18.627	3.505
NT-v2 100M	18.921	-3.663	-4.594	17.999	17.770	4.084	17.038	0.597
NT-v2 250M	22.810	-8.311	-6.730	18.155	18.351	2.350	20.246	-1.061
NT-v2 500M	23.442	2.011	-0.685	21.938	21.963	8.909	22.415	1.558
Hyena tiny-1k d128	6.210	6.746	4.322	7.354	7.920	5.155	5.322	2.697
Hyena tiny-1k d256	7.710	7.390	4.647	8.584	9.063	6.323	6.532	3.920
Hyena tiny-16k	5.355	6.763	4.482	6.506	7.258	4.536	4.775	2.915
Hyena small-32k	7.990	7.127	5.165	8.170	7.742	6.891	7.385	4.209
Hyena medium-160k	7.319	4.327	4.285	6.493	6.405	6.076	6.589	4.000
Hyena medium-450k	8.223	9.629	5.521	9.277	11.258	6.714	7.286	4.523
Hyena large-1m	7.216	7.373	4.592	7.710	9.122	5.768	6.108	4.322
Caduceus-Ph 1k d118	5.570	4.253	4.259	5.821	4.736	5.131	5.533	4.178
Caduceus-Ps 1k d118	7.216	8.032	5.625	8.212	7.034	6.209	6.654	4.179
Caduceus-Ph 1k d256	9.968	1.134	3.655	7.555	4.697	7.909	9.081	7.412
Caduceus-Ps 1k d256	9.564	5.699	4.004	8.436	7.863	7.827	8.620	6.190
Caduceus-Ph 131k	13.178	6.715	5.408	13.211	13.423	10.759	12.266	9.510
Caduceus-Ps 131k	13.935	6.309	3.965	12.520	12.385	9.836	12.857	3.710
Evo 1	7.709	-3.004	-3.280	6.045	8.720	-1.444	7.384	4.290
Evo 2	23.166	-1.725	-1.658	-1.076	10.680	3.036	11.824	18.995

Table 42: Mean coding score of edited **non-coding** sequences, full length, CDS vs intergenic.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	-21.227	0.854	-4.935	-2.577	1.058	-21.940	-22.032	-20.577
DNABERT-2	-20.380	-1.592	-6.445	-1.772	3.020	-20.342	-20.287	-19.344
NT-v2 50M	-19.989	-10.623	-15.801	-12.961	-7.681	-24.793	-19.713	-19.520
NT-v2 100M	-18.687	-8.832	-14.230	-11.007	-6.388	-22.448	-17.994	-18.608
NT-v2 250M	-22.297	-13.754	-16.118	-14.250	-10.007	-24.914	-22.138	-21.927
NT-v2 500M	-23.772	-4.965	-10.989	-7.703	-2.135	-27.426	-23.581	-23.490
Hyena tiny-1k d128	-11.512	-0.760	-3.468	-2.685	1.494	-11.186	-11.183	-11.216
Hyena tiny-1k d256	-12.603	-0.424	-3.867	-2.841	1.625	-12.240	-12.228	-12.392
Hyena tiny-16k	-11.149	-0.514	-2.963	-2.321	1.421	-10.860	-10.841	-10.867
Hyena small-32k	-18.088	0.715	-2.822	-1.779	1.405	-16.550	-16.535	-17.618
Hyena medium-160k	-13.271	-2.354	-3.169	-2.989	-0.017	-12.409	-12.338	-13.098
Hyena medium-450k	-13.616	1.378	-2.977	-2.064	2.557	-12.497	-12.493	-13.408
Hyena large-1m	-12.045	-0.207	-2.559	-1.849	1.851	-11.630	-11.572	-11.962
Caduceus-Ph 1k d118	-11.616	-0.477	-2.352	-1.594	0.151	-11.422	-11.412	-11.440
Caduceus-Ps 1k d118	-13.188	-1.548	-1.973	-1.834	-1.865	-12.645	-12.614	-13.066
Caduceus-Ph 1k d256	-14.758	-0.926	-2.183	-1.247	-3.722	-14.252	-14.251	-14.508
Caduceus-Ps 1k d256	-15.100	1.185	-2.191	-0.925	0.210	-14.482	-14.490	-14.677
Caduceus-Ph 131k	-14.566	2.715	-1.568	-0.196	-0.186	-13.739	-13.746	-14.400
Caduceus-Ps 131k	-14.353	1.033	-3.134	-1.962	-0.687	-13.976	-13.948	-14.350
Evo 1	-7.964	-6.954	-7.214	-7.146	-5.977	-8.117	-8.124	-7.861
Evo 2	-5.854	-3.608	-4.188	-3.814	-3.354	-6.065	-6.061	-5.836

Table 43: Mean coding score of edited **coding** sequences, full length, CDS vs lncRNA.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	5.075	-0.531	1.047	3.600	2.495	2.996	3.504	0.215
DNABERT-2	8.305	-1.626	-1.081	5.476	6.901	1.704	6.433	-3.122
NT-v2 50M	11.259	-6.883	-7.825	9.899	8.644	0.100	9.685	-1.991
NT-v2 100M	9.466	-7.580	-7.361	9.239	9.497	-2.440	8.661	-9.420
NT-v2 250M	12.620	-10.782	-8.160	10.555	11.986	-3.810	12.086	-11.131
NT-v2 500M	12.987	-4.899	-6.212	10.746	9.208	0.792	11.802	-16.930
Hyena tiny-1k d128	2.824	2.173	1.240	3.052	3.498	1.946	2.125	0.960
Hyena tiny-1k d256	3.516	1.774	0.958	3.323	3.620	2.312	2.627	1.150
Hyena tiny-16k	2.792	1.546	1.627	2.670	2.578	1.929	2.139	0.855
Hyena small-32k	3.654	1.480	1.547	3.123	3.043	2.844	3.256	1.533
Hyena medium-160k	3.684	0.869	1.527	2.906	3.102	2.713	3.221	1.736
Hyena medium-450k	3.688	2.266	1.785	3.338	3.131	2.600	3.015	1.515
Hyena large-1m	3.473	2.212	1.376	3.002	3.759	2.502	2.870	1.460
Caduceus-Ph 1k d118	2.888	-0.493	1.070	2.121	0.617	2.130	2.591	1.780
Caduceus-Ps 1k d118	3.082	0.740	1.579	2.896	1.540	2.201	2.629	1.256
Caduceus-Ph 1k d256	3.641	-1.078	0.816	2.935	1.119	2.691	3.221	1.909
Caduceus-Ps 1k d256	4.019	-0.236	1.079	3.350	2.421	2.684	3.225	1.493
Caduceus-Ph 131k	4.270	0.624	0.917	4.100	2.734	3.111	3.850	-0.026
Caduceus-Ps 131k	5.134	-2.606	-0.540	5.054	3.093	2.343	4.713	-3.362
Evo 1	5.292	-8.268	-7.693	1.990	4.654	-4.821	5.180	2.447
Evo 2	6.710	-6.049	-5.083	-5.268	2.353	-3.401	3.522	3.776

Table 44: Mean coding score of edited **non-coding** sequences, full length, CDS vs lncRNA.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	-10.410	-1.036	-1.430	-1.069	-3.049	-10.971	-10.956	-9.962
DNABERT-2	-14.345	-2.832	-4.080	-2.637	-2.263	-14.995	-14.907	-14.655
NT-v2 50M	-11.855	-8.949	-10.185	-9.812	-9.497	-16.802	-12.154	-11.890
NT-v2 100M	-13.331	-8.445	-9.002	-8.712	-8.933	-17.673	-12.526	-13.516
NT-v2 250M	-14.835	-12.551	-10.474	-10.767	-12.285	-18.945	-13.961	-14.695
NT-v2 500M	-15.634	-6.516	-8.230	-8.024	-8.346	-21.011	-15.342	-15.435
Hyena tiny-1k d128	-6.060	1.560	-0.921	-0.016	0.520	-6.073	-6.084	-6.190
Hyena tiny-1k d256	-6.837	0.949	-1.188	-0.340	-0.273	-7.092	-7.103	-6.919
Hyena tiny-16k	-7.106	0.722	-0.707	-0.112	-0.332	-7.107	-7.095	-7.178
Hyena small-32k	-9.010	1.074	-0.381	0.320	-0.561	-8.439	-8.434	-8.683
Hyena medium-160k	-7.931	0.220	-0.443	-0.175	-0.984	-7.535	-7.540	-7.045
Hyena medium-450k	-7.487	1.224	-0.544	-0.056	-0.436	-7.215	-7.194	-7.158
Hyena large-1m	-7.278	1.005	-0.633	-0.182	-0.323	-7.086	-7.083	-6.665
Caduceus-Ph 1k d118	-6.657	-1.168	-0.532	-0.300	-2.289	-6.777	-6.787	-6.424
Caduceus-Ps 1k d118	-8.332	-0.567	-0.105	0.105	-1.999	-8.198	-8.201	-8.317
Caduceus-Ph 1k d256	-8.596	-1.049	-0.377	0.141	-2.992	-8.453	-8.510	-8.322
Caduceus-Ps 1k d256	-9.552	-0.332	-0.366	-0.099	-2.066	-9.377	-9.382	-9.165
Caduceus-Ph 131k	-9.430	0.737	0.078	0.387	-3.512	-8.839	-8.830	-9.078
Caduceus-Ps 131k	-11.189	-2.398	-1.211	-1.346	-5.264	-10.766	-10.784	-10.153
Evo 1	-8.813	-9.825	-8.819	-9.606	-8.592	-9.291	-9.341	-8.765
Evo 2	-5.130	-7.311	-5.860	-6.464	-7.271	-5.423	-5.420	-4.898

Table 45: Mean coding score of edited **coding** sequences, full length, CDS vs intron.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	18.132	5.733	10.241	14.872	12.119	13.920	14.864	8.309
DNABERT-2	21.547	5.731	9.043	20.954	20.269	13.605	18.109	13.669
NT-v2 50M	21.458	-7.211	-3.303	17.727	14.006	6.921	18.843	4.815
NT-v2 100M	20.672	-1.581	-0.842	19.553	19.422	7.686	18.318	4.690
NT-v2 250M	22.944	-6.822	-0.094	17.527	15.735	6.341	20.014	0.580
NT-v2 500M	22.203	0.137	2.165	19.359	15.681	10.439	19.262	8.171
Hyena tiny-1k d128	6.649	4.654	4.863	6.958	6.952	5.353	5.606	2.420
Hyena tiny-1k d256	8.730	3.982	5.626	7.876	7.122	6.799	7.061	3.412
Hyena tiny-16k	5.961	3.769	4.764	5.901	5.496	4.725	4.931	2.139
Hyena small-32k	9.086	4.883	5.901	8.114	7.227	7.688	8.244	4.011
Hyena medium-160k	8.046	3.784	4.948	6.949	7.124	6.529	7.182	4.125
Hyena medium-450k	9.709	7.047	6.885	9.622	9.327	7.768	8.344	5.198
Hyena large-1m	7.879	5.528	4.774	7.119	8.072	6.111	6.514	3.466
Caduceus-Ph 1k d118	6.855	2.758	4.753	6.091	4.550	5.797	6.290	4.119
Caduceus-Ps 1k d118	6.954	3.063	4.796	6.305	5.036	5.735	6.144	3.245
Caduceus-Ph 1k d256	10.781	1.832	4.381	8.185	5.188	8.529	9.691	6.890
Caduceus-Ps 1k d256	11.655	1.532	4.810	8.788	5.695	9.278	10.309	7.042
Caduceus-Ph 131k	13.781	4.529	5.522	12.518	12.469	11.187	12.681	15.582
Caduceus-Ps 131k	15.798	1.965	4.539	12.830	10.142	12.679	14.931	9.858
Evo 1	10.856	-3.920	-4.265	5.766	10.599	-2.045	9.661	10.466
Evo 2	20.519	-3.609	-3.446	-3.177	9.511	2.083	10.721	15.985

Table 46: Mean coding score of edited **non-coding** sequences, full length, CDS vs intron.

model	original	nt shuffle	dinuc Markov	codon shuffle	syn swap	stops in	stops out	rev comp
GROVER	-17.226	-1.274	-4.452	-3.773	-2.818	-18.710	-18.746	-12.407
DNABERT-2	-19.519	-0.806	-7.835	-3.697	0.964	-20.919	-20.900	-15.042
NT-v2 50M	-18.057	-12.459	-14.526	-14.009	-11.488	-22.410	-18.789	-15.161
NT-v2 100M	-17.518	-5.168	-11.500	-9.239	-5.191	-20.509	-17.373	-13.905
NT-v2 250M	-21.239	-10.951	-11.535	-11.371	-10.500	-23.294	-21.401	-17.242
NT-v2 500M	-21.026	-5.760	-9.123	-8.422	-4.520	-24.518	-21.729	-17.591
Hyena tiny-1k d128	-10.356	0.124	-2.792	-2.173	0.637	-10.245	-10.327	-8.194
Hyena tiny-1k d256	-11.559	-1.339	-4.332	-3.606	-1.260	-12.313	-12.410	-8.520
Hyena tiny-16k	-10.148	-1.181	-2.981	-2.653	-0.673	-10.646	-10.674	-7.848
Hyena small-32k	-11.212	0.744	-2.631	-1.710	-0.873	-11.216	-11.287	-8.545
Hyena medium-160k	-11.042	-0.378	-3.109	-2.464	-0.811	-10.901	-10.909	-8.806
Hyena medium-450k	-10.852	2.145	-3.008	-2.029	0.789	-10.905	-10.925	-7.703
Hyena large-1m	-11.113	1.154	-3.062	-2.108	0.567	-11.109	-11.124	-8.708
Caduceus-Ph 1k d118	-8.053	-1.221	-2.047	-1.588	-1.588	-8.449	-8.448	-6.023
Caduceus-Ps 1k d118	-9.342	-2.040	-2.252	-1.851	-2.062	-9.290	-9.324	-7.478
Caduceus-Ph 1k d256	-10.584	-1.760	-2.268	-1.756	-3.173	-10.526	-10.613	-8.139
Caduceus-Ps 1k d256	-11.451	-1.494	-2.534	-1.996	-1.818	-11.263	-11.303	-9.731
Caduceus-Ph 131k	-10.692	1.722	-1.727	-0.560	-0.343	-10.296	-10.342	-9.364
Caduceus-Ps 131k	-11.798	-0.083	-2.237	-1.605	-2.345	-11.484	-11.532	-9.419
Evo 1	-7.946	-7.168	-7.079	-7.312	-6.215	-8.153	-8.087	-6.971
Evo 2	-5.006	-4.891	-4.985	-4.910	-4.798	-5.097	-5.097	-3.598

A.2 Summary tables

These are the per-model summaries written by `compare.py`, restricted to the models in this report. Column names: *emb* = embedding probe, *6mer* = 6-mer probe, *resid* = residual probe, *cod/nc shuf* = fraction of nucleotide-shuffled coding / non-coding sequences called coding, *stop in/out* = fraction of coding sequences called coding after in-frame / out-of-frame stop insertion, *revcomp* = same after reverse complement, *p3/p12* = codon-position ratio, *nons Δ* = mean signed Δ of nonsense mutations in coding sequences.

Table 47: Summary, 200 bp Genomic Benchmarks.

Model	emb	6mer	resid	cod shuf	nc shuf	stop in	stop out	revcomp	p3/p12	nons Δ
GROVER	0.906	0.895	0.592	0.929	0.633	0.868	0.880	0.779	1.022	-0.268
DNABERT-2	0.912	0.895	0.574	0.906	0.555	0.866	0.893	0.787	1.003	-0.393
NT-v2 50M	0.893	0.895	0.567	0.809	0.408	0.785	0.862	0.778	1.002	-0.507
NT-v2 100M	0.886	0.895	0.556	0.837	0.452	0.773	0.850	0.735	0.986	-0.537
NT-v2 250M	0.907	0.895	0.577	0.766	0.357	0.731	0.897	0.800	0.932	-1.202
NT-v2 500M	0.892	0.895	0.557	0.842	0.439	0.798	0.856	0.779	0.986	-0.538
Hyena tiny-1k d128	0.866	0.895	0.539	0.904	0.496	0.823	0.825	0.708	1.024	-0.123
Hyena tiny-1k d256	0.879	0.895	0.547	0.904	0.516	0.837	0.839	0.727	1.012	-0.160
Hyena tiny-16k	0.864	0.895	0.546	0.892	0.448	0.826	0.840	0.724	1.006	-0.097
Hyena small-32k	0.879	0.895	0.561	0.914	0.498	0.856	0.868	0.747	1.007	-0.111
Hyena medium-160k	0.881	0.895	0.564	0.873	0.433	0.836	0.868	0.712	1.022	-0.164
Hyena medium-450k	0.868	0.895	0.570	0.885	0.444	0.822	0.827	0.719	1.022	-0.124
Hyena large-1m	0.873	0.895	0.558	0.873	0.460	0.811	0.831	0.686	1.029	-0.176
Caduceus-Ph 1k d118	0.874	0.896	0.555	0.923	0.509	0.843	0.863	0.741	1.022	-0.091
Caduceus-Ps 1k d118	0.887	0.896	0.563	0.939	0.571	0.865	0.881	0.750	1.011	-0.113
Caduceus-Ph 1k d256	0.900	0.896	0.579	0.934	0.557	0.873	0.896	0.757	1.090	-0.172
Caduceus-Ps 1k d256	0.911	0.896	0.581	0.932	0.576	0.878	0.900	0.772	1.091	-0.230
Caduceus-Ph 131k	0.922	0.896	0.614	0.972	0.671	0.906	0.928	0.858	1.006	-0.307
Caduceus-Ps 131k	0.936	0.896	0.631	0.943	0.606	0.875	0.942	0.782	0.851	-0.783
Evo 1	0.894	0.896	0.611	0.457	0.116	0.530	0.901	0.727	0.904	-1.632
Evo 2	0.931	0.896	0.627	0.620	0.195	0.677	0.878	0.902	0.587	-5.118

Table 48: Summary, full length, CDS vs intergenic.

Model	emb	6mer	resid	cod shuf	nc shuf	stop in	stop out	revcomp	p3/p12	nons Δ
GROVER	0.988	0.987	0.606	0.933	0.570	0.965	0.972	0.894	1.010	-0.298
DNABERT-2	0.994	0.987	0.623	0.817	0.398	0.961	0.987	0.965	0.981	-0.371
NT-v2 50M	0.996	0.987	0.618	0.258	0.026	0.831	0.997	0.719	0.924	-0.976
NT-v2 100M	0.996	0.987	0.609	0.177	0.025	0.744	0.996	0.573	0.880	-0.764
NT-v2 250M	0.998	0.987	0.618	0.076	0.007	0.634	0.995	0.462	0.840	-1.727
NT-v2 500M	0.996	0.987	0.599	0.626	0.199	0.871	0.991	0.643	0.899	-0.880
Hyena tiny-1k d128	0.980	0.987	0.554	0.962	0.449	0.952	0.963	0.778	1.025	-0.084
Hyena tiny-1k d256	0.979	0.987	0.556	0.952	0.460	0.957	0.961	0.813	1.021	-0.107
Hyena tiny-16k	0.977	0.987	0.574	0.965	0.439	0.951	0.960	0.803	1.014	-0.056
Hyena small-32k	0.986	0.987	0.595	0.977	0.598	0.971	0.981	0.856	1.007	-0.081
Hyena medium-160k	0.979	0.987	0.589	0.888	0.297	0.957	0.971	0.867	0.999	-0.082
Hyena medium-450k	0.987	0.987	0.582	0.983	0.614	0.966	0.980	0.878	1.022	-0.103
Hyena large-1m	0.982	0.987	0.597	0.959	0.493	0.959	0.971	0.899	1.014	-0.088
Caduceus-Ph 1k d118	0.973	0.987	0.574	0.969	0.468	0.955	0.969	0.893	1.009	-0.033
Caduceus-Ps 1k d118	0.981	0.987	0.572	0.936	0.318	0.962	0.973	0.838	1.007	-0.081
Caduceus-Ph 1k d256	0.989	0.987	0.582	0.663	0.415	0.972	0.985	0.924	1.100	-0.111
Caduceus-Ps 1k d256	0.993	0.987	0.573	0.968	0.677	0.985	0.990	0.923	1.085	-0.097
Caduceus-Ph 131k	0.993	0.987	0.616	0.965	0.712	0.985	0.989	0.986	0.974	-0.167
Caduceus-Ps 131k	0.996	0.987	0.628	0.975	0.583	0.967	0.993	0.850	0.967	-0.276
Evo 1	0.991	0.987	0.628	0.192	0.017	0.318	0.991	0.900	0.913	-1.528
Evo 2	0.991	0.987	0.613	0.118	0.030	0.705	0.975	0.958	0.584	-2.810

Table 49: Summary, full length, CDS vs lncRNA.

Model	emb	6mer	resid	cod shuf	nc shuf	stop in	stop out	revcomp	p3/p12	nons Δ
GROVER	0.953	0.932	0.605	0.431	0.374	0.845	0.876	0.538	1.011	-0.176
DNABERT-2	0.968	0.932	0.656	0.379	0.293	0.641	0.920	0.261	0.955	-0.434
NT-v2 50M	0.993	0.932	0.651	0.074	0.024	0.492	0.980	0.354	0.947	-0.747
NT-v2 100M	0.991	0.932	0.662	0.033	0.018	0.312	0.987	0.299	0.873	-0.742
NT-v2 250M	0.993	0.932	0.657	0.017	0.003	0.202	0.993	0.068	0.859	-1.376
NT-v2 500M	0.991	0.932	0.644	0.170	0.087	0.537	0.985	0.156	0.903	-0.793
Hyena tiny-1k d128	0.927	0.932	0.560	0.884	0.782	0.856	0.878	0.696	1.011	-0.064
Hyena tiny-1k d256	0.937	0.932	0.558	0.769	0.679	0.859	0.891	0.686	1.030	-0.086
Hyena tiny-16k	0.922	0.932	0.572	0.833	0.708	0.857	0.893	0.681	1.011	-0.056
Hyena small-32k	0.956	0.932	0.593	0.791	0.725	0.911	0.933	0.768	1.001	-0.051
Hyena medium-160k	0.949	0.932	0.618	0.666	0.563	0.906	0.936	0.792	1.004	-0.060
Hyena medium-450k	0.942	0.932	0.621	0.841	0.722	0.892	0.913	0.758	1.021	-0.079
Hyena large-1m	0.946	0.932	0.620	0.810	0.671	0.892	0.916	0.757	1.001	-0.054
Caduceus-Ph 1k d118	0.936	0.932	0.574	0.368	0.255	0.884	0.924	0.852	1.020	-0.057
Caduceus-Ps 1k d118	0.948	0.932	0.579	0.642	0.406	0.908	0.944	0.796	1.024	-0.062
Caduceus-Ph 1k d256	0.958	0.932	0.589	0.336	0.309	0.917	0.947	0.858	1.025	-0.070
Caduceus-Ps 1k d256	0.969	0.932	0.599	0.474	0.441	0.930	0.954	0.769	1.063	-0.090
Caduceus-Ph 131k	0.971	0.932	0.654	0.639	0.650	0.935	0.965	0.571	0.979	-0.085
Caduceus-Ps 131k	0.983	0.932	0.655	0.099	0.117	0.848	0.981	0.135	0.980	-0.216
Evo 1	0.987	0.932	0.673	0.005	0.004	0.081	0.984	0.857	0.958	-1.144
Evo 2	0.983	0.932	0.664	0.007	0.000	0.057	0.936	0.922	0.615	-2.049

Table 50: Summary, full length, CDS vs intron.

Model	emb	6mer	resid	cod shuf	nc shuf	stop in	stop out	revcomp	p3/p12	nons Δ
GROVER	0.990	0.995	0.596	0.831	0.426	0.977	0.980	0.892	0.994	-0.350
DNABERT-2	0.996	0.995	0.648	0.748	0.434	0.959	0.984	0.962	0.977	-0.493
NT-v2 50M	0.998	0.995	0.629	0.066	0.005	0.875	0.990	0.826	0.902	-0.947
NT-v2 100M	0.997	0.995	0.622	0.360	0.160	0.904	0.995	0.798	0.888	-0.683
NT-v2 250M	0.999	0.995	0.630	0.072	0.006	0.840	0.997	0.584	0.854	-1.325
NT-v2 500M	0.997	0.995	0.625	0.511	0.173	0.926	0.993	0.887	0.913	-0.679
Hyena tiny-1k d128	0.987	0.995	0.549	0.932	0.572	0.969	0.975	0.802	1.007	-0.112
Hyena tiny-1k d256	0.988	0.995	0.544	0.861	0.444	0.966	0.973	0.851	0.993	-0.156
Hyena tiny-16k	0.979	0.995	0.560	0.904	0.431	0.965	0.969	0.788	1.004	-0.091
Hyena small-32k	0.989	0.995	0.569	0.931	0.603	0.977	0.982	0.882	1.001	-0.110
Hyena medium-160k	0.987	0.995	0.581	0.877	0.511	0.974	0.983	0.892	1.008	-0.112
Hyena medium-450k	0.992	0.995	0.595	0.953	0.688	0.971	0.975	0.903	1.018	-0.139
Hyena large-1m	0.989	0.995	0.603	0.942	0.633	0.964	0.975	0.847	1.010	-0.113
Caduceus-Ph 1k d118	0.986	0.995	0.543	0.837	0.356	0.968	0.979	0.940	1.008	-0.090
Caduceus-Ps 1k d118	0.990	0.995	0.556	0.821	0.267	0.975	0.984	0.886	1.002	-0.099
Caduceus-Ph 1k d256	0.992	0.995	0.571	0.734	0.282	0.984	0.986	0.934	1.063	-0.133
Caduceus-Ps 1k d256	0.994	0.995	0.585	0.700	0.334	0.991	0.991	0.929	1.089	-0.143
Caduceus-Ph 131k	0.995	0.995	0.626	0.864	0.666	0.985	0.991	0.983	0.948	-0.172
Caduceus-Ps 131k	0.997	0.995	0.639	0.663	0.488	0.976	0.995	0.967	1.010	-0.250
Evo 1	0.996	0.995	0.648	0.066	0.011	0.228	0.991	0.969	0.950	-1.613
Evo 2	0.995	0.995	0.643	0.009	0.001	0.678	0.974	0.931	0.587	-2.250

A.3 Reproduction

From `Model Evaluation/Coding-Noncoding/Embedding-Patterns/`:

```
# 200 bp analyses for one model (default environment; DNABERT-2 and NT-v2 use -e legacy)
GLM_MODEL=PoetschLab/GROVER pixi run python kmer_baseline.py
GLM_MODEL=PoetschLab/GROVER pixi run python perturb.py
GLM_MODEL=PoetschLab/GROVER pixi run python codon_position.py

# Full-length data (downloads listed in the script go to data/full/raw/)
pixi run python prepare_full_length.py
GLM_DATASET=full ./run_models.sh <gpu> <env> <model ids...>   # runs A, B and C for each model

# Summaries and Experiment D (cached embeddings only; no GPU needed)
pixi run python compare.py
pixi run python noncoding_signatures.py
pixi run python noncoding_signatures_full.py

# This report
pixi run python report/make_tables.py && typst compile report/report.typ
```